

REPORT ON PROCEEDINGS BEFORE

PUBLIC ACCOUNTABILITY AND WORKS COMMITTEE

INQUIRY INTO DATA CENTRES

CORRECTED

At Macquarie Room, Parliament House, Sydney on Friday 8 May 2026

The Committee met at 9:20 am

PRESENT

Ms Abigail Boyd (Chair)
The Hon. Scott Farlow (Deputy Chair)
The Hon. Jacqui Munro
The Hon. Emily Suvaal

PRESENT VIA VIDEOCONFERENCE

The Hon. Mark Buttigieg
The Hon. Taylor Martin
The Hon. Peter Primrose

The CHAIR: Welcome to the second hearing of the Committee's inquiry into data centres. We are starting ever so slightly late due to tech issues. I am going to resist the urge to make jokes about that. I acknowledge the Gadigal people of the Eora nation, the traditional custodians of the lands on which we are meeting today. I pay my respects to Elders past and present, and celebrate the diversity of Aboriginal peoples and their ongoing cultures and connections to the lands and waters of New South Wales. I also acknowledge and pay my respect to any Aboriginal and Torres Strait Islander people joining us today or joining us online. My name is Abigail Boyd, and I am the Chair of this Committee.

I ask everyone in the room to please turn their mobile phones to silent. Parliamentary privilege applies to witnesses in relation to the evidence they give today. However, it does not apply to what witnesses say outside of the hearing. I urge witnesses to be careful about making comments to the media or to others after completing their evidence. In addition, the Legislative Council has adopted rules to provide procedural fairness for inquiry participants. I encourage Committee members and witnesses to be mindful of these procedures.

Dr BRONWYN CUMBO, Lecturer, University of Technology Sydney, affirmed and examined

Professor KURT IVESON, Professor, University of Sydney, affirmed and examined

Dr RIKI SCANLAN, Postdoctoral Research Fellow, University of Sydney, affirmed and examined

The CHAIR: I welcome our first panel of witnesses. Thank you so much for giving your time to give evidence today. Would you like to start by making a short opening statement?

BRONWYN CUMBO: Chair and members of the Committee, thank you for the opportunity to appear today and for initiating this important inquiry into data centres. I think we can all agree that data centres are necessary infrastructure. They enable so many digital and AI services that our modern society has become reliant upon, and they will enable many future innovations that we cannot yet imagine. To me, this is not about whether we should or should not have data centres; this is about how we have data centres. How can we shape what is a global industry and a globally networked infrastructure in a way that aligns with and supports the things that matter to the Australian people and our environments?

Sydney's data centre industry, as it looks today, has not been shaped by intentional design; it has been shaped by default. We have all been caught out by the number of data centre applications, the pace at which the industry has entered the market and clustered across our industrial lands, and their increasing size and resource intensity. This inquiry provides a valuable opportunity to pause and consider what the industry could look like if it was designed around the values and priorities of the Australian people—things like having access to affordable housing; having water and energy security; living in a thriving, climate-resilient community; keeping our environments healthy and biodiverse; and maintaining a healthy economy and diverse employment opportunities. Here, we have the opportunity to interrogate how the industry is intersecting with these core priorities and how it can be designed to support us to meet them and minimise any cost to them.

I put forward six recommendations in my submission that reflect some of my thinking on this issue. It is based on research I've been carrying out for the last year and a half through the University of Technology Sydney. These six recommendations call for more strategic, inclusive and long-term approaches to how we sight, assess and govern this industry. They centre on three core issues: Firstly, integrating data centres thoughtfully into our urban and regional fabric, so they align with and actively support other key priorities; secondly, updating how we assess data centres, so they're recognised as the transformative infrastructure they are and so the cumulative impacts of clusters on communities and environments are properly captured; and, thirdly, bringing local communities meaningfully into the process, because right now the people living alongside these facilities have very little influence, but they're the ones bearing the costs. I welcome the opportunity to provide evidence.

KURT IVESON: I'll do an opening statement on behalf of both myself and Dr Scanlan. I want to join Dr Cumbo in thanking you very much for the invitation and for your work on the issue. I almost just want to say what she said. I think we have very similar perspectives on many of these issues. In my opening statement, I'll just pick up on the idea of data centres as infrastructure, which I think is the thing that also informs the submission that we made to the Committee. Indeed, they are. I work at a university, and it's hard to imagine how we could operate without access to the computation and data storage provided by data centres. It's hard to imagine how the Committee would operate, notwithstanding the hassles this morning, but that doesn't justify further hasty expansion of data centres or provide an argument against careful planning and regulation. Indeed, we think the opposite is true. By definition, infrastructure requires strategic planning.

We don't let anyone just set up a new bus route, build a new road, open a new hospital or start a new school without rigorous assessment against a range of public interest standards. Transport, health and education infrastructure are way too important to be left to the individual decisions of market actors. The same is true for communication and, we think, is becoming true for computation as well. In developing infrastructure, government plays a critical role in planning and coordination. And it's only through good government that we can ensure that individual data centres then end up serving the public interest by meeting standards, as Dr Cumbo has pointed out also, that maximise community benefit and minimise community and environmental harm and contribute to the broader policy objectives that we've set.

This is all the more important, I think, because data centres are also privately held assets in AI supply chains. They're being built in haste as part of the accumulation strategies of some of the largest corporations and investment firms on the planet, who are seeking to stimulate and capture the market in AI applications and then to profit from that. It's our Government's job to then represent the public interest here, to disentangle public interests from private interests and to plan accordingly. We're in the early days of research, and I guess the project that we're working on around communications infrastructures—we've more been at the point of reviewing the

literatures around this and so thought that we'd share some of the insights from that literature with the Committee, but at a different stage of our research than Dr Cumbo.

But the recommendations we've got focus on developing planning principles which centre public interest by prioritising net zero ambitions in determining the energy demands and efficiency of data centres, ensuring their transparency in operation, mitigating the potential for resource diversion from other pressing priorities like housing, minimising community and environmental impacts, and insisting on social licence through community engagement. I guess we'd finish by saying we think we should be ambitious here. There's an opportunity actually, with the Committee's work and with the Government consultation underway right now, to establish New South Wales as a site of regional and global best practice that can avoid some of the problems that have occurred elsewhere and set benchmarks for other jurisdictions. Thanks very much.

The CHAIR: Thank you very much. I'll start by asking some questions, and then I'll throw it open to my colleagues, some of or most of whom are online at this point. But they'll be in and out as we go. Both of you in your opening statements and in your submissions have come to the nub of what my concern was in relation to these data centres. We can all agree we need some data centres. The question is how many do we need, where do they need to be, how do they need to be built and a bunch of other things. But one of the issues is that we seem to have quite a lot of secrecy around what each of these data centres are being used for and what they are hosting. On that basis, I note, Dr Cumbo, you recommend that we need a framework to distinguish between what is a critical data centre infrastructure development and what is non-critical. Given that operators like AirTrunk don't actually know what their hyperscale clients are storing, how can we do that? What would that look like in terms of knowing which is critical and which is not critical?

BRONWYN CUMBO: I can't speak to the legal elements, because a lot of—the secrecy is not intentional. It's by default design, in a way. Their governance systems with co-locators have—like you say, AirTrunk has many clients, and they don't know what data they're housing inside their data centres. I don't know how we can create a differentiation. But, if there's some kind of legal mechanism—we don't need to know what data they're housing, but maybe data categorisation is a kind of future speculations. We're not saying AWS is hosting data from Commonwealth Bank. We're saying what is critical and what is not critical, and there will likely be a spectrum of criticality, and part of the motivation for this is around things like climate resilience. The industry's very—it has a lot of redundancy built in. There's usually two or three copies of the same data allocated across access zones that are geographically disparate. They can be in different access zones in Sydney—or Sydney and Melbourne, or Australia, overseas—and that is partly to ensure resiliency if a data centre does go down.

But in terms of the materiality and the environmental cost, the assumption there is that we have this abundance model, that it doesn't really matter what the material costs are to the data centres on the ground. Looking at what is critical from a material perspective, through this kind of model of criticality, could enable more climate resilience. During floods, if we do need to shut down data centres, if we have a drought and low levels of water availability, these kind of future scenarios where digital access comes up against people's access to potable water—these kind of more future scenarios, speculations can help us think through what is necessary in terms of the different types of data.

The CHAIR: Can we tease that out? Can we bring Professor Iveson and Dr Scanlan in on this as well? For example, you've got a number of data centres being built in Lane Cove. We've got this cluster there with a lot of community concern about that. You could have data centres that are purely hosting things like Netflix and you could have data centres that are actually being used to store government information, but we don't know which is which. The community presumably would want to know when they are working out, at that initial planning stage, whether it is worth them having the impacts on their natural environment and amenities et cetera, in order to justify that particular project.

But as you said, Dr Cumbo, in an emergency scenario when decisions have to be made—in a flood or a drought or whatever—on which ones we turn off, at least the Government would want to know what is being hosted there, presumably. Do you think that there needs to be a way for the Government, at least, to have more intervention or more power over what is in these data centres to begin with? Sorry, that was a very long question. I'm trying to tease that out.

BRONWYN CUMBO: From a material perspective, yes—if we're looking at the environmental impacts of these things. But I also think there needs to be a critical interrogation about what is critical—Netflix would think Netflix data is critical—depending on these different kinds of things in terms of how we're ranking and valuing the different data functions for our society. We have low literacy in terms of what data we use, how data works and the interaction between these different types of data that enable our society to function the way it does.

Often it is very interrelated, so I think pulling those different things out—the other thing is that data doesn't just reside in a data centre; it moves around. Identifying critical and non-critical data inside a data centre is also a

challenge in terms of the architecture of the infrastructure, because it resides there and moves. It's mobile, which I think Data Centres Australia mentioned last week. It doesn't just live in this house. I'm not sure if that answers your question well enough, Abigail.

The CHAIR: It does. What's the counter to that? Would these data centres argue that if we knew which ones were critical and at what level of criticality it would make them a security threat? What is the industry counter to that?

KURT IVESON: If I understand it correctly, that would be one of the counters. The other would be—as I think you've just put it, Bronwyn—that there is an abundance model. Innovation comes with cloud and AI computing with no limits. I think where the discussion has to be at is that we just can't have that, because it's so energy hungry that there are going to be limits. There are limits in land, there are limits in energy and there are limits in computer chips. There are limits, so arguing that there should also be conscious limits because of our decarbonisation and our net zero emission strategies adds weight to that.

I think, exactly as you say, the difficulty is we could all sit here and take the technicality out of it for a moment, and we could actually imagine a spectrum of computing, storage and processing that is essential through to trivial and think that if we're going to have limits, we would want to prioritise the essential over the trivial. If government and communities are making that distinction, I have no answer to you about how that can technically be implemented. I'm not a computer engineer. We need the industry to take that desire of ours seriously and figure out how to deliver it, which is an easy thing for me to sit here and say. That is where the discussion has to be at.

The CHAIR: This is a great discussion, because that idea of data centre development has been built so far. It has come out of this abundance model, as you say. I think people have got used to the idea of data not having a physical footprint, so most people are surprised when you say we've almost got 100 data centres already in New South Wales. It's a physical manifestation of what you've been seeing happen over the last few years in AI and ChatGPT and everything else. I think that's coming as a surprise to people.

Because the community hasn't really been aware of that and we're only just finding out really, as a community, that these things are incredibly data hungry and water hungry and could set back our emissions targets and all of those things, do you think there's an argument, as an immediate plan, to say, "Even if you can't tell us what these things are for and we can't tease out what's critical and not critical, if we put some very strict limits on what you can do in terms of how much water you can take and how much energy you can take, then at least the industry would develop in a way that reduces those impacts while understanding itself what all of those things are being used for"? Does that make sense? If they can't tell us what they're used for, would putting these kind of constraints on force them to innovate and make those decisions themselves?

RIKI SCANLAN: Decisions about criticality? I think that, from a data centre perspective, criticality means profit. This is the difficulty that you run into. I think government is going to be very capable at compiling a list of where they have data, what data it is, what is critical from the Government's standpoint—health data, particularly, for example—so we can have a policy which says, "We know what data we have, and you have to make sure that that's first cab off the rank." Going further runs into, yes, I guess, commercial confidence or security issues with the companies themselves. That spectrum of criticality to essential to trivial becomes very difficult when it comes to private applications, some of which are very important and necessary functioning within society, some of which are less. Those judgements are difficult for data centres to reliably make. This, I think, speaks to the need of just greater government engagement and responsibility in this area.

BRONWYN CUMBO: Can I just pick up on that slightly? I think what you're talking about, Chair, is setting limits, the material limits, and allowing the data centre industry to sort itself out in terms of what they think is most important. If that's the right interpretation—

The CHAIR: Sort of. Yes, whether they would innovate, in the same way that DeepSeek created a more efficient model, apparently because of the result of the restriction on the GPUs that were allowed to be sold to China. If we were to proactively put limits on them, would there at least be some curbing of that abundance culture?

BRONWYN CUMBO: I definitely think that would be the case. They would say that they'll go elsewhere, to other jurisdictions, where they haven't got those limitations placed on them, so there is that commercial threat that the industry will put forward. In terms of going back to what we're doing here and our values as people, I think the material limits really do talk to people—the environments. The extractive nature of the industry just by design means that we do need some kind of policy regulation around how much they take. How we design that, I think, is where the integration and the opportunity could sit. The data centre industry is not necessarily a bad industry. It could offer so many opportunities if carefully integrated, like Professor Iveson mentioned—the integration and strategic planning and that part is really missing at the moment. So how can we strategically plan by emphasising

that this is a material industry? Then I think there will be different forms of innovation. You can't say it's a speculative industry, but—yes.

The Hon. SCOTT FARLOW: I've got to say, Professor Iveson, I was very surprised, in reading your submission, to learn that Sydney was number six in terms of data centre concentration in the world.

KURT IVESON: Yes.

The Hon. SCOTT FARLOW: That was something that I didn't appreciate before, in terms of the amount of data centres that exist within Sydney. How has that come to be? What's been attractive about Sydney to lead to us being number six in the world when it comes to data centres?

KURT IVESON: Like you, I was actually a bit surprised. The way that I understand CBRE did those calculations was about energy capacity. So it's not about just the number of data centres, but about the energy capacity. This is I guess why we're having the inquiry. It's been a bit haphazard. There's a range of things that seem to drive that geography. If you look at the global submarine sub-sea cable map, Sydney is very well connected compared to other places in Australia, certainly, and also just well connected in South-East Asia. There's land, there's energy, there's geopolitical stability, there's a skilled labour force, there are institutions like ours and a knowledge economy here, and I think probably also the outcome of the way in which Sydney's positioning as a global city that has corporate headquarters and large clients with regional operations here. Those are the reasons it's attractive for the people whose computing is happening inside those data centres to be here, and hence driving that demand for data centres.

If I may, just to say, this is one of the things that I think is the opportunity of the policymaking that we can do right now, which is that you can't just pick up and move to Darwin. I mean, there are data centres being built in Darwin et cetera, but there are advantages that Sydney has that won't just magically disappear overnight. If we do some planning, and do some strategic planning, those advantages will still be here. There are clustering and agglomeration effects that are embedded in the urban economy and the urban environment here that suggest that the threats that we'll just pick up our operations and go somewhere else if you try to change any of the settings I think are a bit hollow.

The Hon. SCOTT FARLOW: You also suggested that Australia reportedly had the second highest investment pipeline when it came to data centres as well. I am interested in where that "reportedly" supposedly comes from, in a sense, in terms of that second highest investment pipeline. What does that mean? What does that look like in comparison to other countries?

KURT IVESON: Let me take that on notice. I can't remember without spending five minutes fishing through the footnotes of the report. From memory, that was one of the industry presentations at a recent Committee for Sydney event that was co-hosted with Endeavour Energy, which I'll circulate afterwards, if that's okay. But those two things, the CBRE thing about what we've currently got and then the statistics about the investment pipeline, do suggest that—I mean, you could look at that CBRE thing and still see that the United States graphs, or their numbers, are still way higher.

The Hon. SCOTT FARLOW: Looking at that—for instance, somewhere like Atlanta—you're looking across a set of different cities. It seems like relatively consistent growth across a range of areas, but one of them I was interested in was Atlanta, which had a huge spike in interest. What was the change in policy settings in somewhere like Atlanta to encourage that kind of growth and investment?

KURT IVESON: That, I can't answer. I'm sorry.

The Hon. SCOTT FARLOW: No worries. Dr Cumbo, in terms of the data centre location, in your submission I think you outlined that some of the concerns were that, effectively, there is a cost to the neighbours, so to speak, when it comes to data centres—I think we've seen that throughout Sydney as well—and potentially in poor planning results too. At somewhere close to me, Macquarie Park, for instance, we have a very well connected metro line in the area. We've seen a proliferation of data centres, when arguably there would be better uses for either high-employment activities that occur there or residential development because of something like the metro line. How do you think we could potentially have what we all agree we need in terms of data centres, but located in areas where there is less impact on neighbours or less detriment to, potentially, employment or residential housing growth?

BRONWYN CUMBO: I think the location question to me seems like a planning question right now. As far as I understand, data centre developers buy industrial land, and there was land available in Macquarie Park. Once you have one data centre, they will cluster. The same has happened in Lane Cove. This is where it has just happened by default. In terms of trying to mitigate that, looking at the planning gaps and different types of zoning—industrial and employment lands have always kind of just been let be. Local councils have their different

zones that they put out but they rely on diversity just happening. So this monopoly that's occurring has caught all the councils by surprise.

I think looking at the planning instruments and how we can reconsider different zonings. There's also a need to really think about how we're defining and categorising data centres. Currently, they fit across a number of different types of zones—business enterprise, light industrial, I don't know all the different types of zoning. That's similar to a warehouse but these are hugely resource-intensive, noisy, disruptive infrastructures and they don't just come alone. Thinking about the cumulative impacts as well is something that's not currently built into the planning system, particularly environmental and social impacts.

For instance, if a diesel generator—if there is a power outage in a cluster of data centres, what is the cumulative air impact going to be from nitrous oxide outputs? That's not currently assessed in the planning documents. They're assessed as individual assets.¹ It is the same with cumulative impacts on the local environment. The cluster in Lane Cove is right next Lane Cove River and bushland. Over time, how is that going to shift and shape that bushland area, which is very important to the local people? Also the sonic impacts on residents living next to these very loud data centres. There's already a lot of research coming out in terms of the impact on health over time.

The Hon. SCOTT FARLOW: To that point, how loud are data centres?

BRONWYN CUMBO: I couldn't give you an exact number. I was reading yesterday that they are high decibels in terms of impact on hearing if you're located close by. I'll take that on notice.

RIKI SCANLAN: We have a figure on this, which is approximately 40 to 59 decibels. If you're in a decently populated restaurant and there's people conversing around you, it could be a loud conversation about you—is the usual sort of comparison figure. One thing to note about this is that the nature of the noise is different from the pleasant bubble of a restaurant conversation. It's a drone and there are distinct, subjective qualities to the nature of the sound, which is important to take into account. Just to get into the technicals of the planning framework, the planning frameworks don't take into account the subjective perception of sound. They just take into account the decibel level. If I can also come back on the second largest investment pipeline, I have the source from that: a 2025 Knight Frank report called *Global Data Centres Reports*. I think the figure there is actually for Australia at large—

The Hon. SCOTT FARLOW: Not just Sydney.

RIKI SCANLAN: Not just Sydney in particular—although there is data for individual markets by city in that report.

The Hon. SCOTT FARLOW: With respect to the diesel generators, how often are we seeing with data centres that they use diesel generation as backup from usual electricity? Is there a percentage that's allocated?

BRONWYN CUMBO: They are turned on every few months, as I've been told, for testing. They get turned on when there's a sag in the grid or there's the power outage. Not all data centres have only diesel generators as backup power; some have batteries. But the number of diesel generators inside data centres is quite substantial. I can't give you exact numbers in terms of, for instance, a 110 megawatt data centre and how many diesel generators they'd have, but there are a lot. When you're regularly testing those generators every few months, there are EPA guidelines to keep that individual data centre below the air-quality limits. I'm not sure if the cumulative impacts of multiple in a cluster are effectively assessed in the planning process framework at the moment.

The CHAIR: For example, that one on Mars Road in Lane Cove, they're saying that proposal has 49 backup diesel generators. If there's a cluster, you're saying that Planning hasn't really looked at the impact of if there was a lag in the grid that suddenly you would have all of them turned on at the same time?

BRONWYN CUMBO: Yes, and it's not monitored. Speak out if I'm saying anything or you've read something differently, but it's part of the lack of environmental monitoring of data centres.

The CHAIR: And dealing with them all as individuals rather than as a—

BRONWYN CUMBO: A collective. They're very much interconnected.

The Hon. SCOTT FARLOW: Just to that point, because I've heard this discussion before, so to speak, when this is raised and it's been indicated to me, "Well, if you were to compare other uses that could occur in an

¹ In correspondence to the committee dated 30 June 2026, Dr Bronwyn Cumbo, Lecturer, University of Technology [clarified](#) their evidence.

area like that, for instance a warehouse, the amount of truck movements you'd have per day." Is there any sort of comparative as to what the impact of that potential diesel generation would be compared to if you had a warehouse in the same facility and the amount of truck movements that would have on a daily basis?

BRONWYN CUMBO: Not that I've read. I will say data centres run 24/7. I think that's one of the main differences between normal light industrial, which, for instance, the Mars Road area accounted for. There's a lot of residential land right by it, so having that 24/7 noise, light as well—they are all lit up—does have this longer constant impact.

The CHAIR: Professor Iveson and Dr Scanlan, in your submission you talk about the idea of every new data centre facility being matched with additional renewable generation from the day it's open. Is that achievable in New South Wales? What would we need to do to get there?

KURT IVESON: You can feel me hesitating because we've got colleagues that I know you're talking to later today, indeed, who are more energy specialists than we are. For us it was more sort of zooming out and saying, "Well, if we can't do that, should we be making the approvals in the first place?" In other words, what's our priority? If it's not possible, and if it looks as though further growth in an already crowded data centre sector here is going to undercut and undermine our net zero ambitions, then which of those two things is our number one priority and, frankly, if we can't achieve it, should we be making any further approvals, is kind of the question. You could take the Ireland approach, which is to not insist that the renewables are online at the moment the data centre opens, but to give a ramp-up period of a few years where you can at least make a credible demonstration that you've become the anchor investor in a new renewable energy facility, and we're going to start now but in two years that solar farm is going to be operational. The Irish approach is to take that into consideration.

The CHAIR: I think we've come back full circle to the beginning, about a lot of this depends on whether, as a society, when we're weighing up the cost of this, what they're being used for. As an example, in the Southern Highlands there's a massive hyperscale data centre being proposed there that has its own gas generator that they're proposing, which would be as big as Kurri Kurri. That will have a really large impact on our total ability to cut our emissions and meet our climate targets, but no-one can tell us what that will be used for, and they claim there's quite a lot of secrecy around that use. How do we work out how to deal with that when we don't know what it's for in the first place? Do we just need more transparency over the industry?

KURT IVESON: My guess would be that the governance regimes that seem to be emerging in other jurisdictions are kind of a combination of what you're talking about, which is on the one hand to actually specify what the energy allocations are going to be for any further data centre developments, and only make a certain amount available and require anybody that's applying for access to those allocations to demonstrate what they're doing on a range of issues. Not just about energy, not just about whether it's renewable or not, but to make some stipulations about energy efficiency, community benefit and minimising environmental harm.

None of them that I'm aware of have also then gone this extra step of making a stipulation around critical or non-critical functionality. Just reflecting on the earlier conversation we had, as you say, we'll go round and round on that, in a way. I think that's where possibly putting our thinking about data centres together with some of the discussions that we're having about regulating AI and its applications is also part of that thing. This is all part of a big supply chain that data centres are one link in. We can be thinking about how we regulate data centres to actually prioritise non-trivial and critical uses, but we can also think about how we're regulating computation and applications of computing in different parts of society to try to minimise the trivial functions.

To give you one micro example of that, we're both members of our union on campus. We're in enterprise agreement negotiations right now and we have clauses about the use of AI in the workplace that are seeking to say we don't just want a free-for-all where we're literally setting up bots to answer student queries potentially, say, 24/7 if that's going to have a material footprint that, when you talk to students about that, they're horrified with. There's a society-wide conversation that we can be having that has to extend to data centre planning, but also to uses and applications of the technology, which again seems like a big ask, and these things are moving so quickly, but I think that's where we need to be at.

The CHAIR: Are they two separate discussions or are they the same discussion?

KURT IVESON: No, I feel like they're the same discussion, coming at it from two different sides. That's how I would see it. To achieve what I think we're all talking about—about saying this isn't limitless, there are limits and we have to now make priorities given those limits—one way we can attack that is from the infrastructure side of things and from the data centre side of things. And I think it's incredibly important that we do so. Another line of intervention is on the application and the software side of things, and also just thinking about how we intervene in what's acceptable and what's not acceptable, what's critical and what's not critical now as well.

The CHAIR: One of the big points of disagreement between the submissions we've had from the data centres versus submissions we've not had, because we haven't had a government submission, but we've had statements coming out from Sydney Water and from AEMO and others that are very different in terms of the amount of resources that they anticipate data centres will use in New South Wales. For example, 20-odd per cent of Sydney Water's total usage is the utility's anticipation versus 1.9 per cent from the data centres. It is the same with energy efficiency. Could we circumvent all of that by saying, "Fine, okay, then over the next 10 years we will allow you to have 2 per cent of entire water usage and you can then bid for that." Presumably that would encourage innovation and competition and perhaps sidetrack the whole debate in some ways. Is that what we should be doing?

KURT IVESON: Riki may want to pick up on this as well. It's very early days to know what the impact of that kind of approach has been in Singapore. But I guess what we can say, from the two calls for applications just in the energy space at the moment, for instance, where they've literally just said, "We're making 200 megawatts available of capacity for data centres. Apply, and we will set some baseline criteria. About 50 per cent of the energy for your data centre has to be renewable. You have to be making X contributions to Singapore's innovation economy", et cetera. You can certainly see that that has driven innovation in energy efficiency there. It's driven innovation in cooling. Singapore is a hot place and the data centres that have been applying through those calls for applications have literally been inventing new ways to cool things, allowing temperatures to get a little bit hotter but still maintaining efficiency. That approach of limiting resources to drive innovation—early days, I think—seems to be driving the kind of innovation that you're talking about in Singapore.

BRONWYN CUMBO: Could I say something on that? Because it's not just water and energy, it's also land. It's also entangled all this stuff. Finding these locations and capping resourcing is one way you could approach it, because then you can coordinate the social impact, the environmental impact, and they can work together. So it will require a bit of collaboration between the industry. If they are clustered in that particular area, how can they maximise efficiencies between each other, if that's something that is possible. That's another kind of consideration in terms of the material spatial quality of this. It would probably make it easier for utility providers. Sydney Water, because they work in spatial catchment zones, they can actually think about water-based infrastructure and innovate with the data centre industry. The same with transmission lines and this kind of relationship with the REZ. These types of material relationships are very spatial. So thinking about how they can work together around that, which is what's happened in Singapore, it is inherently spatial in the way they've designed it.

KURT IVESON: You're sounding like a geographer there, but anyway.

RIKI SCANLAN: I can come in on a technical point regarding these numbers and forecasts that are available. As you point out, there are a variety of numbers. Just to give some, I guess, guidance on the interpretation or the reliability of the variety of these numbers, I would say that AEMO's numbers in general are highly reliable. The forecasting that they have is clear. It's rigorous. There's a methodology that's outlined in detail about various factors going forward, different elements of different models for the short or the long run. In a word, this is reliable stuff.

At the very other end of the spectrum are what you might describe as back-of-the-envelope calculations. Here I'm referring to the Mandala numbers in particular, which were referred to last week, which are sourced not from ABS water accounts. That's for the total water use, which is the denominator and their figures. The numerator just covers their sources, ultimately—an ABC article—which then comes back from Sydney Water in origin. If we're talking about whether we trust Sydney Water's 90 gigalitres figure in the future, we have to also think about that they are the source of numbers for current water usage at present. I think there are important questions to be asked about whether the 90 gigalitre figure is accurate. Is the 2 per cent figure going to be accurate? There are going to be serious questions that we need to answer first if we want to set a limit and what limit that might be. But we need to make sure that the actual methodology underlying these things are rigorous calculations, not just numbers drawn out of thin air.

The CHAIR: How are they so wildly different then? Twenty-five per cent versus 1.9 is a massive difference. Is the Mandala one just not credible? You're being very polite, but is that the—

RIKI SCANLAN: It's hard to say, I think, honestly. There could be technical change in how the industry operates over the next few years. What they've done is essentially take the current energy figures, the 2030 energy figures from AEMO, the current water figures, which come from the ABC article—so from Sydney Water, ultimately—and multiply the two together to get some number for 2030. That's an estimate. It goes as far as it goes. It doesn't tell you something about the underlying demand structure. It doesn't tell you anything about the underlying capital investment that's playing out in the sector—in other words, all the detailed work that AEMO has done on the electricity data. It could be that that number fits into the story. It's hard to say. The other point

here is that these are average figures calculated for an annual basis, which then don't take into account fluctuations in water demand, which then have implications on the daily water use of particular communities. It doesn't take into account the costings that then may play out in terms of water bills down the line. There's a whole range of issues here that I think we need to get right before we proceed with mass approvals.

The CHAIR: Is that something the Government should be doing, ideally? Is that something where the Government could say to its departments to go and actually do that work? That would seem like a logical first step. Yes? You're nodding, for Hansard.

RIKI SCANLAN: Yes.

KURT IVESON: For the record, yes.

The Hon. SCOTT FARLOW: I might just pick up on the energy thing and then I'll hand back to you. In terms of that modelling going forward, there is, of course, a financial incentive for lower energy and water use for data centres. What kind of development are you seeing in that space in terms of lowering energy use and lowering water use in the development of data centres globally?

KURT IVESON: Again, with apologies, because some of this is going to be anecdotal from me, but reading about, for instance, some of the big players in Australia—I know CDC, for instance, are quite proud of their water efficiency achievements over the years that have been driven in their account partly by the fact that they were sort of born in a context of drought in inland Australia. How I think that should be interacting with regulation is that if those standards can be achieved—

The Hon. SCOTT FARLOW: Why can't everyone do it?

KURT IVESON: —then why can't everybody do it? Shouldn't we be demanding that everybody lift to that standard? Certainly, we absolutely mustn't be punishing people who are lifting to those standards by allowing others to come through and undercut—that are less efficient and more resource intensive—because we're not thinking about resource allocations or pricing the resources effectively.

BRONWYN CUMBO: I wanted to just mention something on that in terms of energy efficiency. The industry is doing a lot in terms of investing in renewable energies. How that plays out on the ground is the murky part, because there's already a backlog of renewable energy projects—how the data centre industry is fitting into, for instance, EnergyCo's landscape with the REZ. It does require people to build the infrastructure. You need to have relationships with communities to be able to lay these things out. It's a slow process of negotiation, and there are lots of reasons why it will be held up. All the industry can do at the moment is invest. That's the avenue for them. It is this system that requires a lot of different levers to be pulled for it to become a renewable system. In terms of what the industry can and is doing, that's kind of the limit of what they can do, as I understand right now, through their power-purchasing agreements.

The CHAIR: Can I ask you, Dr Cumbo, in your research, you've looked at the community inclusion in decision-making elements and made recommendations around that. I know from going out and speaking to people in Lane Cove and in the Southern Highlands and other places that they are very angry. It's a combination of the secrecy, that these things are primarily being fast-tracked without the opportunity for community consultation. It strikes me that we're not going to bring the community on board with even critical data centre infrastructure unless we begin to include community. Can you talk us through what your recommendations are?

BRONWYN CUMBO: I will just pull up that particular point from my submission. There are a few things in terms of community participation. Firstly, getting them on board early on—so bringing councils in and looking at how they've zoned their industrial lands and looking at how many data centres they could potentially house, because at the moment industrial lands are just being congregated by data centres. Having more council input into that planning process at the beginning and then looking at where they're located in relation to communities. I really think it's such a waste for these data centres to be located so close to residents so it's going to impact on their lifestyle and their everyday lived experience of living in a place.

There are industrial lands that are located further away intentionally because of the noise and pollution impacts. Is there a way of thinking about that, also considering the equity issues around that? Naturally in the inner city, there's not as much space as somewhere like Western Sydney. It does require strategic planning around how we can make this fair. In terms of the community burdens, making visible a lot of the costs and labour that communities are carrying out as a consequence of the industry. There is a lot that local councils are doing to try and catch up and that the community is doing to try and speak up. Because they don't have opportunities right now, it's coming across as resistance rather than participation in an inclusive process.

More broadly, in terms of how we imagine the industry, bringing communities into that process. So if we are reimagining what the data industry could look like for New South Wales—what do the public think and does

it align with their values. That's more of a speculative future. This industry will change. These big boxes might become smaller. There's distributed models people have been talking about, if we have things like autonomous vehicles in cities, we're going to need smaller little data centres around. Different people I've spoken to in the industry have different forecasts in terms of how the industry will shape and how it will be integrated with our urban and regional fabric. That's something to consider in terms of how it plays out.

KURT IVESON: One of the little lines that stuck with me from an article by Alex Johnson is about the distinction between the resources associated with data centres flowing to a community or flowing through a community and just passing right on out the other side. That helpfully just gets us to be thinking about—if we're thinking about data centres as physical manifestations of computation that have to be in a place, they have to be somewhere, and somebody's going to be living close or not too far away from them if they're in an urban environment, and then just taking seriously those questions and not just saying, "The community benefit is X trillion dollars of economic productivity uplift".

That is not really something that can be attributable to our local data centre, but actually really thinking about the local community, as well as the broader picture. There are places, for instance, right now in Eastern Creek and Kemps Creek, even if we think about the industrial areas, where there are massive data centres existing there already and in construction, incredible fibre-optic connectivity being built to those places, and 10 minutes up the road in communities that I'm working with on other issues, in Bidwill and around Mount Druitt, where connectivity is terrible.

How are we thinking about the co-benefits where, if all this investment is happening to connect the data centre, for instance, it could also be enhancing the surrounding community's internet connectivity? If there's waste heat and people are cold in winter, because they can't afford to turn on their heaters, how could we be thinking about how a data centre could be making a contribution by heating people's homes? All these things are possible technically. If we were strategically planning and engaging communities in those kinds of conversations—again, there are people in the industry who are really actively taking this stuff seriously and trying to think about what being a good neighbour would look like. So let's encourage it, let's require it and let's reward it when it's there, rather than just leaving it to chance for it to happen.

The CHAIR: Just looking at those claims from the—well, I call it the AI real estate industry, but from the data centre industry, I think that Mandala report said that they were going to have almost 18,000 people working in data centres by 2030. There has been a lot made of this being an incredible economic benefit to New South Wales. What do you make of those claims?

KURT IVESON: Come on, mate.

The Hon. SCOTT FARLOW: Here's something I prepared earlier?

RIKI SCANLAN: Yes, we discussed some of these numbers. Their figure is 17,900 by 2030. But just to be clear, total New South Wales employment at present is about 4½ million, so this would be currently about 0.2 per cent. The data centre industry likes talking about their energy numbers being less than 1 per cent; their employment numbers are also less than 1 per cent. In the strategic planning for exactly these Western Sydney industrial areas, there are employment targets which are calculated on a per-hectare basis in order to support the growth of residents and jobs in Western Sydney, as important targets. Data centres fall well below the target average for these areas. They know that. These are highly capital-intensive and not labour-intensive workforces and sectors. The employment effects—it's a growth, but we also have to take into account that these are data centres that are substituting from the existing data centres in Commonwealth Bank's basements.

So if you've got three people employed to manage a Commonwealth Bank data centre who are now being laid off because Commonwealth Bank has got rid of it, and now it's one-quarter of a person for the Commonwealth Bank data centre operation in a large hyperscale or co-locating data centre somewhere else, perhaps there's actually a net drop in employment overall. They like to talk about how their data centres support employment figures that are far larger than these, as well, but you have to remember that this is about the overall tech industry growth, which is not something that is directly reliant in a directly causal way. The growth of data centres, in other words, doesn't cause employment in software engineering. I think we can safely say that the employment effects in terms of data centres are relatively small, if not negative.

The CHAIR: In terms of that investment quantification, it seems to be based on how much in terms of construction materials are purchased, what the labour hire is during that period of construction. How much of that is actually of benefit to the New South Wales economy? I'm thinking about the idea that they're importing products from overseas. How much of it is actually not then just going into the pockets of these data centre companies?

RIKI SCANLAN: In terms of the construction?

The CHAIR: Yes. When we talk and they say, "There's going to be this boost to productivity and this boost to GSP," what is that merely in terms of—

RIKI SCANLAN: I'd say there are two different parts of the story here. One is the operational productivity side of things, which I'll come to second, and the other is the construction side of things. That's the easiest to deal with. We know that we have a construction skills shortage and we have a construction materials shortage. The question here is also a resource allocation question about whether or not we're going to meet our housing targets and build housing, or we're going to build data centres, or we're going to build the kinds of industry that we need to support employment in places like Western Sydney.

Here the narrative sometimes is that data centres are specialised construction projects that have specialised resources and inputs and skills. This is true to the extent that there are genuine differences with the ways in which these things are built. But then it's also very much not true when it comes down to "We've got a greenfield site out in Western Sydney. We need to clear the ground. We need to lay in the concrete foundations. We need to put in the steel." These are generic construction jobs that constitute the majority of the labour time and the resources that go into the construction of these sites. And that is a question about diverting from other possible construction uses that we have going on right now. So I wouldn't say that it's going to be a boost to the construction sector. Maybe there are going to be some effects within construction, but we are talking ultimately here about an allocation problem. The overall constraints in the construction sector mean that we've got to raise serious questions on that side of things.

The CHAIR: Sorry, can we just delve into that and then I will let you do the other side? We often hear now about not just the pressures on finding labour and construction materials for big housing projects, but also whenever we're talking about transport projects we talk about blowout costs because suddenly it's more expensive for labour and suddenly it's more expensive for materials. If we are instead approving data centres, is that also pushing up the costs of other public projects and detracting from the speed at which they can be done?

RIKI SCANLAN: This is essentially, yes, a demand and supply story. When the demanding side here of data centres and the real estate developers want to pay a large sum of money to get something done very quickly in order to meet global market demand on their own side, this will push up prices in the short run.

The CHAIR: Okay, and you were going to say the other side, so carry on.

RIKI SCANLAN: Absolutely. On the other side of the productivity story there are large unknowns. The large unknowns are about AI. I'm not confident about making predictions in that space, but there are a million predictions here that spread in every direction, whether it's going to be total collapse of employment and the economy as we know it through to mild adjustments that are going to be positive. So this is difficult. In terms of data storage and computing, it becomes a little clearer, although I think it's still quite difficult because, for most applications, where your data is or where your computing is taking place matters a bit in terms of pricing, in terms of access, in terms of speeds. But if you're downloading material from a cloud program because you're an engineer or an architect and you've got three or four gigabytes worth of material that you're downloading to work from your personal workstation and your office is in Sydney, that it takes 20 seconds longer or 30 seconds longer because you're downloading it from Singapore versus Sydney seems to me to be a pretty marginal productivity effect for that individual worker as treating the data centres as an input into downstream productivity.

You've got to take into account what is the counterfactual story: If we didn't build this many data centres and we build just the ones that we really need for immediate use in Sydney for government, for critical infrastructure, following that point from earlier, what would be the effects on productivity in that case? This is hard to say. You'd have to do quite complicated calculations and production functions to calculate what exactly is going to go on in a story, but we can't just simply say this is the value of the industry and that's therefore the productivity that we are enhancing over and above the alternative if we don't do it. You actually need to do quite serious analysis, which is not being done at present.

The CHAIR: And look at the opportunity costs and look at the—

RIKI SCANLAN: Precisely.

The CHAIR: I think this needs to be the last question. Apologies, because we were running late. If you look at somewhere like Macquarie Park, where there have been warnings that the connections to the data centres and the construction there have delayed some connections to new housing. The City of Ryde council has raised concerns about data centres reducing the long-term employment density in the area. Is what's happening in Macquarie Park a kind of case study in how data centre clustering, if done poorly, can undermine the broader planning objectives in a region? Would you view that as being a bit of a case study for what we maybe don't want to repeat in the future?

RIKI SCANLAN: I think that captures it.

BRONWYN CUMBO: I can talk a bit to that one. Yes, is the short answer to that. I think it's also a question of scale and where we're looking. The data centre industry tends to talk in terms of a nationwide unit of scale, and then you look at what's going on on the ground and the impacts are uneven. Macquarie Park is a case study of an uneven cost that's borne by the community. That is one example. But it also shows the lack of strategic planning, because Ryde council had plans for housing and the land was bought up by data centre developers. Everyone was working within the rules, and that's just how it emerged. That will continue without some kind of intervention around it, because the industry is looking for space—infrastructure—to continue to build there.

The CHAIR: I have one final question. Late last year, the Government set up the Investment Delivery Authority to fast-track a bunch of projects. That included a bunch of data centre projects which are now online. Do you think that was a mistake, going down the wrong track to further fast-track these projects rather than slowing them down?

BRONWYN CUMBO: I'm curious as to how the IDA is contributing to fast-tracking, because they still have to go through the DA process. One thing I did notice—and correct me if I'm wrong—is that there are a lot of Australian-owned companies that are listed. I think that might have an impact on economic benefit in terms of taxation. But I don't really know how the IDA is contributing to speeding up things at the moment. I don't know if you have a clear idea of that. I haven't been able to find any information around how it's fast-tracking in actuality yet.

The CHAIR: It seems a bit conflicting.

The Hon. SCOTT FARLOW: I think there are a lot of questions as to whether it is or not.

The CHAIR: A lot of claims made and a lot of—

The Hon. SCOTT FARLOW: It might be more spin than substance.

The CHAIR: We could talk to you all day, but you have other things to do and we have other witnesses to hear from. Thank you so much for your time. Apologies again for the late start. To the extent that there were questions taken on notice or there are supplementary questions, the Committee secretariat will be in touch. That concludes our time with you today.

(The witnesses withdrew.)

Mr ANDREW SJOQUIST, Chief Executive Officer, WinDC, sworn and examined

The CHAIR: We now welcome our next witness. Would you like to start by making a short opening statement?

ANDREW SJOQUIST: Certainly. Chair and members of the Committee, thank you for the opportunity to appear today. My name is Andrew Sjoquist, and I am a founder of WinDC. From our perspective, this inquiry is really about how New South Wales and Australia position themselves for the next wave of AI and digital infrastructure in a global race. Artificial intelligence demand is growing extraordinarily quickly, and globally we are seeing infrastructure development timelines, whether grid, transmission or planning, struggling to keep pace with the speed of technology deployment that is evolving on a monthly basis, whilst infrastructure often evolves on a year or decade basis. It creates a major opportunity for New South Wales if we think differently about infrastructure and regional development.

At its core, a data centre converts electrons into photons. Electricity goes in and digital services come out. But around that sits a much broader ecosystem: grid, substations, batteries, fibre, cooling systems, construction, maintenance and skilled operational work. These are long-life infrastructure assets that can help support regional economies for decades if planned properly. Water is often described as the source of life, and rightly so. Communities and industries depend on it. That is why at WinDC we have pursued waterless and low-water cooling approaches, because we believe future infrastructure must work alongside the needs of regional communities, agriculture and local environments.

Historically, data centres have concentrated around Sydney because that was where electricity demand, population of users and connectivity were centred, but Australia's energy systems have changed and our users are now worldwide. Today increasing amounts of generation infrastructure sits in regional New South Wales through solar, wind and storage projects. At the same time, large volumes of renewable energy are effectively being wasted because the grid cannot always absorb or transport it effectively.

In 2025 alone, around 7.2 terawatt hours of renewable energy was curtailed or spilled in regional Australia. That is enough energy to supply roughly two million homes and more than double the estimated electricity consumption of Australia's entire built data centre sector today. I think that highlights the scale of the opportunity for regional New South Wales. What WinDC has explored is an additional infrastructure model, placing appropriate forms of compute closer to energy resources where it makes technical and economic sense. Importantly, this is not about replacing metropolitan data centres. There needs to be a mix of infrastructure models—different horses for courses.

Some applications require low-latency metro facilities; others, particularly large-scale AI workloads, may well be suited to regional locations aligned with energy infrastructure. Done properly, this can create highly skilled regional jobs, support local contractors and suppliers, strengthen regional infrastructure investment, and allow regional communities to participate directly in the growth of the digital economy, not simply watch it concentrate back to the capital cities. WinDC is proud to partner with Yirigaa, a First Nations organisation developing the skills of the next generation in regional and remote Australia. Australia has a genuine opportunity not only to support domestic demand but to participate in a rapidly growing global AI infrastructure market—a huge export opportunity for the State. Meeting that opportunity will require new thinking around infrastructure, energy integration and planning frameworks. I look forward to your questions.

The CHAIR: Thank you very much, Mr Sjoquist. Thank you for your submission. I found it incredibly useful. To be honest, we haven't heard much from many people in the industry wanting to speak directly to us, so it is incredibly useful. One of the largest concerns that first got me interested in what was happening from a planning perspective with data centres was in relation to this increasing demand on the energy grid at a time when we were trying to transition away from our reliance on fossil fuels. If you've got a target to be moving away and to be using a rapidly increasing percentage of your energy from renewables, but then you increase your demand, that's going to make that task harder and put that 100 per cent renewable target in danger.

I had preliminary conversations with people where they were telling me that, actually, this could be an opportunity to not just harness bits of renewable energy that really aren't be used or that are being lost because of the time it takes to get them from where they are, but also there is an opportunity to partner with new renewable energy projects as anchor users, if you like, in order to improve the financial viability of new renewables projects. Can you talk to me about what your recommendations are or your experience of how feasible that is as a model to actually advance the development of renewables and even make it quicker?

ANDREW SJOQUIST: Certainly. Maybe a little bit about my background and experience in renewables: I founded another company about 25 years ago. Today we deliver services to over 20 per cent of all renewable

energy assets in Australia. We connect them and secure them from a cybersecurity and connectivity perspective. We're well experienced in this space. We've seen that market evolve quite rapidly over the last 15 years. There has been some conflation around PPAs. They've been leant on quite heavily as an instrument for justification about investments in renewables.

The CHAIR: Sorry, explain that for Hansard.

ANDREW SJOQUIST: Sure. PPAs are power-purchase agreements. It's a forward commitment by an offtaker of electricity to purchase power over the lifetime of an asset. These are very important to the financing of renewable energy assets. However, they don't necessarily solve some of the physics problems. Whilst they're a commercial instrument and an economic instrument, they don't necessarily ensure that the electricity is available at the right time at the right place. This is where the need for transmission to move the electron or the electricity from the place it's generated to the place it's consumed also fits into that mix. A PPA doesn't do anything necessarily to address grid transmission constraints, because it's there to fund the generation asset, not the grid transmission asset.

The CHAIR: Can you tease that out for us? What does that look like physically? Say you're in the Hunter renewable energy zone and you've got a new solar farm being built. What is the benefit of putting a data centre nearby?

ANDREW SJOQUIST: The key benefit from a physics perspective is that you don't have to move the electricity very far. The by-product or the side effect of that is that you don't then lose the electricity. When you move electricity, you do lose it.

The CHAIR: How much are we talking about? I note that we've got Mr Buttigieg online, who is our resident MP with experience in understanding—

The Hon. SCOTT FARLOW: Sparky.

The CHAIR: He's our sparky. He understands this stuff, but the rest of us do not understand it to the same degree. What are we talking about in terms of the loss as it is being moved?

ANDREW SJOQUIST: Electricity is measured by the amount of electricity that reaches the Sydney metro area, the measurement points in Western Sydney. The further you are from Sydney, the greater your loss. There are other reasons why you get increased losses too, like constrained or not enough capacity in poles and wires, for example. The loss is called MLF, margin loss factor. It's somewhere between 20 per cent and 0 per cent. For every generation asset, AEMO knows what that MLF factor is. It's part of the settlement process. If your asset is close to Sydney, you may have a margin loss factor of a handful of percentages; if you're going out further, say northern New South Wales, for example, which is particularly constrained, those margin loss factors are up to 20 per cent of generation lost. Just in the process of sending the power from northern New South Wales to Sydney, if you were to generate a megawatt of power, maybe only 800 kilowatts would arrive in Sydney.

The CHAIR: So you could have an energy hungry data centre that was close by to a new renewables project, and we would then benefit as a society, as opposed to if it was further away from the renewable energy source, because it would reduce that MLF for that particular user? Is that where we're going with that?

ANDREW SJOQUIST: That's correct.

The CHAIR: Saying there's a 20 per cent loss, is there some sort of mathematical way to say a data centre put in a particular place would actually then have an impact on net demand to a far lower degree than what is being anticipated otherwise?

ANDREW SJOQUIST: There would be a calculation process you could do. Thinking through this in a relatively simple way, if you were a data centre that purchased power-purchase agreements from a particular asset and a particular location, you would find out what that asset's MLF losses are and then you would have to apply that as an overhead to your electricity consumption. If you were consuming a megawatt but you were buying your PPA from an asset that had a 20 per cent margin loss factor, for example, you would actually need to be generating 1.2 megawatts to get your resulting megawatt. This is where a lot of data centres have been focused on the power use effectiveness, the PUE measure, which is how effective they are at turning electricity into compute, or into useable power for compute.

But that only measures the impact from when it passes through the perimeter of their property to when it gets used by the compute. It doesn't necessarily take into account all the losses that have occurred along the way from the generation point. Losses occur from the transmission, like I've mentioned. You get it in the transformers as you step up and down voltages. That generally ends up as heat. If you looked at the system holistically, which I think is what we need to do—we need to zoom out a little bit and consider the amount of losses and inefficiencies

that exist system wide—then location, in my opinion, is the fastest way to resolve that. Locating your demand and your load where you're generating your power is the fastest way to remove any loss.

The CHAIR: We were talking with the previous witnesses about this kind of abundance mindset of the industry. I'm attracted to the idea that we should at least be locating some of these data centres near renewable projects. I think there seems to be quite a strong argument for that. I guess I worry that would detract from the incentive to then be more energy efficient as a data centre operator. What's the counter to that, or how could we control that?

ANDREW SJOQUIST: I think data centre operators always want to improve their efficiencies. If you take a commercial mindset to this, the reality of being more efficient is, effectively, you need to purchase and pay for less electricity. There is a commercial benefit in being more efficient, other than just the community and social benefits of it being the right thing to do and reducing waste. How would you do that? I think you just need to simply include all of the system losses in the considerations of facilities' effectiveness and efficiencies. I heard you talk earlier around how to measure that and how to ensure that people are driving those efficiencies. I think you just need to increase the scope of the efficiency so that it incorporates the entire system and not just what happens within the building.

The CHAIR: Can you explain what you mean by that?

ANDREW SJOQUIST: With the power usage effectiveness measure, because it only measures the effectiveness of power within a building, there are various ways that you can improve the efficiency. But if you've drawn that power from a location that is not efficient in the first place, you could be offsetting all the efficiency gains by virtue of the fact that the losses occurred out in the grid, and the argument for that would be that was someone else's problem. But with this abundant mindset, we can't as a society say that this is someone else's problem. It's all of our problem, so a system-wide approach to this needs to be taken rather than just giving a particular building a tick of approval and then pretending that the problem is someone else's outside of the perimeter of your property to deal with.

The CHAIR: If we were to adopt the approach I understand is being used in Singapore, where we say as an industry in New South Wales, "There is this amount of energy and this percentage of energy use that you're allowed to bid for or that you can seek," but we then did that in a sophisticated way, to say, "We will also take into account the total demand that can be made on the grid and things such as your contribution to building renewables and also how close you are to the energy source." What's your view on that idea, of having almost like a market or an allocation of energy that can be used by the industry as a whole and getting the industry to try to innovate to fit within that the best it can?

ANDREW SJOQUIST: I think the industry is already trying to achieve that; there are mechanisms to do that. The Australian Energy Market Operator and the Australian Energy Market Commission do this to some degree. I think the biggest constraint is the grid, purely because of the time it takes to make grid augmentation, to build it and construct it. I don't necessarily think that the abundance mindset is the wrong thing to have. I just think it needs to be applied in a slightly different way. The grid and resources for moving electricity into metropolitan areas needs to be prioritised to people and place, because people need place to live. As for giant hotels for computers, it often doesn't really matter where they are, to some degree.

Those grid projects that do take a long period of time—a decade or more—just can't scale at the speed that AI and the digital economy needs to move at. So it's not that we shouldn't try, but we shouldn't displace people and people's needs over that industrial requirement. If we accept that there's a baseline of things that need to occur—that there needs to be base infrastructure for powering and heating people's homes, or keeping refrigeration systems running—and that these loads are an additional load, then I think it becomes obvious that the only way to avoid the grid is to not use the grid, and so co-location is key.

The second part about this, too, is there's been a big focus on generation over the last decade or two, and even further, since the national energy market formed in 1999. But I don't think there's been a lot of focus on demand management. There's a lot of demand that enters the market that is not efficient; it is wasteful and not particularly well managed. So AEMO again has a market for that, the demand response market, where basically participants can bid their demand into a pool that AEMO, the market operator, can turn on and off as needed. The biggest participant in that in New South Wales is Tomago, the smelter. They are a demand-response participant that can be turned on and off in various conditions.

If a 40-year-old smelter can coordinate themselves, I would imagine a modern AI workload could also do so. That has been done around the world. It's starting to become a real trend where these data centres are becoming energy-integrated assets, so they're able to respond to times of high congestion. What that ultimately means is that

we don't necessarily need to overbuild as much generation or overbuild as much grid, because we're able to more effectively flatten out the demand and align it to generation times.

The Hon. SCOTT FARLOW: I pick up on that point, because one of the things we are often told is that data centres need continuous electricity and continuous operation. That's the need for diesel-generated backups and the like. How does that work in terms of the model where you can scale up and scale down with data centres?

ANDREW SJOQUIST: That is a conception that has been formed, that data centres need 100 per cent of their power 100 per cent of the time. The loads within data centres are flexible. If we start on the facility loads themselves, in terms of the cooling systems, obviously there isn't as much power needed to cool a facility in the middle of winter at night than there is in the middle of summer during the day. That is an obvious overhead. As you move into the workload stacks, in terms of what's actually running in these facilities, traditionally the environments have needed to be up 100 per cent of the time. Those mission-critical systems that run hospitals and banks and airlines will continue, and they do need to be available 100 per cent of the time. But the growth is coming from AI workloads. If we look at the AI workload patterns, they swing wildly in terms of usage. There are very large amounts of fluctuation from a usage perspective.

But what needs to be taken into account there is, overall, your utilisation is only somewhere around about 50 per cent or maybe 60 per cent. If you are to average utilisation at 50 per cent to 60 per cent then, effectively, you've got a 40 per cent or 50 per cent overhead that you've had to build anyway. You've built that capacity just in case it's needed, but it often sits idle, so you end up in this scenario where some capacity is reserved in the infrastructure for those events, but the event doesn't always occur. It's a contingency event, effectively. There is plenty of capability to coordinate those loads, I believe, but it does require the users of the compute and the users of the demand to participate in that. That's not necessarily the data centres themselves; it's the tenants within the data centres that are the ones that control when, how and how often those workloads run.

The Hon. SCOTT FARLOW: That needs, effectively, industry collaboration in being able to do that. But, from your point there, it's not just the hosts; it's the end users that need to collaborate in that market somewhat as well.

ANDREW SJOQUIST: Yes, that's correct but, again, that collaboration can often be incentivised by more effective pricing. We see it in the energy market. You can buy power in the middle of the day now for cents per kilowatt hour. It's just a supply and demand function. If you have a computing function that you're happy to have delayed by a couple of hours because it's not critical, then you will be rewarded with a lower price. Over time, it effectively flattens out the demand. That's what grids like. They like stable, flat demand.

The Hon. SCOTT FARLOW: With respect to your location model, closer to the grid makes a lot of sense when it comes to energy, but the other challenges are in terms of water connectivity and communications infrastructure as well. How do you see that being aligned in terms of where energy generation is occurring in the regions but also being able to get the water infrastructure in location and also the communications infrastructure?

ANDREW SJOQUIST: I'll start with the water piece first, because it's probably the most critical and the most important to community. We've taken the approach that we are going waterless with WinDC facilities. We do that by effectively using closed-loop liquid cooling systems, not dissimilar to how a radiator might work in your car. For us, that hasn't necessarily been a big consideration. The downside of that is that we will use slightly more electricity from a power-usage effectiveness perspective. That's just a physics thing. The heat has to be rejected somehow, so if it's not going to be rejected by evaporating water, we need to reject it by turning more fans, effectively, which takes power.

There is a pro and a con, but we think that preserving the water is probably more important than consuming a little bit of extra power, particularly where we have a situation where we're not incurring MLF losses anyway. On the connectivity front, Australia has invested for the last 20, 30 years huge amounts in fibre-optic cable systems that run all over the country. Telstra has just finished a big build up and down the eastern seaboard, and it's continuing. There are other private enterprises that have built overseas submarine cable systems that come in and connect our capital cities. As we get out into the regions, those systems also penetrate into the regions as well. There are multiple private enterprises that run cables out there.

The Hon. SCOTT FARLOW: How closely do they align with the REZ or renewable energy infrastructure?

ANDREW SJOQUIST: The final builder and owner of communications systems is actually Transgrid themselves. The TNSP has a fibre optic network that's used for their protection relay switching—it's what actually controls the switches at the remote substations. Those assets are going right into the REZ. They're built as part of the high-voltage transmission cables, so it suggests—

The Hon. SCOTT FARLOW: So everything's aligned?

ANDREW SJOQUIST: Very aligned, yes.

The Hon. SCOTT FARLOW: In terms of your model, what current impediments are there for you in terms of the regulatory or planning environment in New South Wales that make it difficult?

ANDREW SJOQUIST: The key one for us is—there's a couple. On the power side, it's about ensuring that we can get fast, effective access to grid studies. Whilst we have a lower demand on the grid, we still need to be mindful that the generation asset also has other obligations to the grid to provide the power that's needed back in the capital cities. In terms of regulation in local and State government, we are keen not to necessarily follow a State significant development process. We believe that we want to engage more tightly with communities. There's a number of things going for us in that stance, because we're fairly low profile. We're containerised, so most of our height is about 12 feet off the ground, that's about it. We can basically disguise ourselves inside of a renewable energy asset that already exists. We also have a closer working relationship with those communities as well. Certainly the discussions that we've already had to date with a number of councils have been very positive. We want to empower them to work with us to make local decisions rather than having to deal with the consequences of a State significant development decision.

The Hon. SCOTT FARLOW: I notice from your website your deployment time is 90 days. That includes all planning pathways and that as well, does it?

ANDREW SJOQUIST: No. Planning pathways, of course, need to be there, but from the time from which we press the go button, if you like, knowing that we're putting extra megawatts of these units in place, to the time that they're commissioned is 90 days. We effectively manufacture them on a supply chain. That supply chain is in the process of being relocated into Australia. Then once they come off that chain, they're fully certified, ready for use. It's very fast compared to building a traditional data centre building.

The Hon. SCOTT FARLOW: How have you found community acceptance so far in terms of your placement in regional communities and your social licence, so to speak, in terms of operations?

ANDREW SJOQUIST: We think it's been pretty positive. It's early days, certainly, for us, but we've had good engagement. I think communities are responding well to being involved in the process. I know there's been a lot of conjecture in some metropolitan suburbs and areas around data centres. Looking at that objectively, part of that may be a bit of a fear of the unknown. Data centres—I've operated them for two decades. I don't necessarily care to go inside another one. I think there's an element here where people often, when they don't know what they are or what they're about, turn to the worst possible thought process around what's occurring there. There's an element of the industry which could do better at educating and perhaps showing people what occurs and just reinforcing the importance. This inquiry relies on data centres right now for us and continues to 24 hours a day, so they are important facilities, but I think they are cloaked in a bit of secrecy. Perhaps now that they're becoming more mainstream and front of mind for everybody, perhaps we can lift the veil a little bit and demonstrate to people why these things are important, as opposed to just keeping them as secret as they are.

The Hon. SCOTT FARLOW: With respect to the model of location closer to renewable energy zones—your model is a modular one. Could traditional data centres or the hyperscalers use a similar model, or are there impediments to them being able to establish in those sort of areas, apart from the zoning, effectively? But if you were to be able to zone either industrial or warehouse zones along some of those areas, would they be able to use a similar model or are there certain impediments?

ANDREW SJOQUIST: I believe they could, and certainly there are some smaller organisations that have done that. Part of the challenge is certainty in policy and making sure that their key ingredients are going to be continued to be delivered well into the future. Traditional data centres are 30 years in terms of their life span, so they need to have some certainty around that sort of length of time, including power supply, people supply—all the ingredients that they need. If you look overseas—US is a good example. I know it's continually said that we're not like the US market, but, again, we are in a global market, so I think we can draw some learnings there. They've been building facilities like this in Virginia, Ohio—Midwest America—for decades.

The Hon. SCOTT FARLOW: Not in the centre of New York and Chicago?

ANDREW SJOQUIST: Yes, exactly right. There's something to be said about that model working there. The Australian market has taken a different approach over 20 years, but maybe there's some rethinking that needs to be done. But certainly they do not want for a 30-year asset to be white elephant by perhaps some changing power policies or other policies that are not necessarily committed for that 30-year time frame.

The Hon. MARK BUTTIGIEG: Thank you for appearing, Mr Sjoquist. It's very, very helpful, as the Chair pointed out before, having a subject matter expert who's down in the weeds dealing with this stuff all the

time. I just wanted to ask you about this trade-off between having—I mean, the logical answer is it's horses for courses. As you pointed out before, there has got to be a mix of both. I wanted to ask about the trade-off—and whether or not people have run calculations on this—between having the data centres located in the renewable energy zones but away from where the main load centres are.

If I were to play devil's advocate, and let's call the devils the electricity distributors, they would argue that, "Look, in terms of latency"—and we heard evidence from previous witnesses about the necessity to have data centres located close to the load density in order to speed up your transmission of information via fibre-optic and how that was important—but they would also say, wouldn't they, to distributors, "Look, there's a place for them to be located out in the REZs, but we've also got to put them close to our dense load centres because with the economies of scale, using the existing traditional electricity infrastructure, we can then use things like community batteries and local solar and wind to cross-subsidise residential customers bills by having that electricity consumption come off the grid, if you know what I mean. There seems to be a balancing act there, isn't there?

ANDREW SJOQUIST: I think it gets into a scenario where you swallow the spider to catch the fly, then you just keep building layers upon layers of things, and the complexity goes up. In its simplest physical form, the latency argument, for one, is one that has been tested now, and it doesn't hold up as well as it used to. I can hear you perfectly well in this conversation, and the latency between us would be probably tens of milliseconds via two or three data centres. The human brain can't necessarily detect latency until you get to about 150 milliseconds of delay—that's when you start to notice the lag. System integrations will notice latency faster, but, again, they're usually operating within the same physical facility, so the latency is again not a consideration there.

In terms of where these things should be placed, certainly the renewable energy zones provide a good opportunity for that. There's a lot of generation happening in one location. I've had a view for a little while that perhaps there should be a DEZ that goes with each REZ—a digital economic zone—for these types of assets. It would also remove or reduce the need for as much transmission necessarily to be built in the backhaul as well. There's been some community conjecture around, "How do we get the power back to metropolitan areas coming out of these renewable energy zones?" Perhaps if we didn't—if we moved the demand closer or the load closer—we wouldn't need to at least move that electricity back into the metro area. I think there's a lot to be said about that, but I think the latency argument particularly is wearing very thin, to be honest.

The Hon. MARK BUTTIGIEG: That's interesting because there has been a lot made of latency in previous evidence. In terms of your particular niche with water and the closed loop system you articulated, can you walk us through the difference between how that works compared to a traditional model of water usage without your closed loop system, which is presumably a set quantum of water that just gets re-used all the time? As you pointed out, it's a bit more electricity intensive because presumably the cooling effect is not as great because you're using a closed loop system. Walk us through the alternative or, for want of a better word, the traditional system. What happens to the water in a traditional data centre with a non-closed-loop system?

ANDREW SJOQUIST: Certainly. In an evaporative cooling system—again, I'm not an expert in cooling systems particularly, but, if you'd like, I'm happy to take it on notice to provide one of our team members who is. But, as I understand it, effectively you have heat that's moved from inside the facility. It goes outside into a cooling tower, and the process of evaporating water around the pipes is what causes the cooling effect. That evaporation process provides the cooling, but, at the same time, of course, the water disappears into vapour into the atmosphere.

The Hon. MARK BUTTIGIEG: I see, so you're not wasting it in the closed loop system. You're saving on water, but potentially increasing electricity.

ANDREW SJOQUIST: That's correct. Just to clarify, in the closed loop system, it's not water in the pipes; it's a refrigerant. It's a little bit more efficient than water at moving heat. But, again, it remains closed loop. It goes through a series of fans, and then the fans blow the heat off the refrigerant pipes.

The Hon. MARK BUTTIGIEG: If you're pumping renewables into that closed loop system, then your electricity consumption thing goes away to a degree, right?

ANDREW SJOQUIST: Again, if we're consuming electricity on the generator side of the grid, then we're avoiding the MLF. Let's assume that we're avoiding, say, 10 per cent of losses on average. It would only increase our power usage by maybe 1 per cent or 2 per cent. It's not significant, but still net advantaged over bringing it all the way to a metropolitan area and the losses that incur in there.

The Hon. MARK BUTTIGIEG: From your perspective, is there a problem in terms of coordinating all this? You spoke a couple of times about zooming out and looking at the bigger picture. It seems to me as though there's a multiplicity of ministries and departments and industry all feeding into this. Do you see a need for an overall coordinating authority, given the prevalence and the uptake of data centres, so that we do this in a smart

way which benefits the whole community, rather than just a market-based sectoral approach and see what happens? Or do you think it will evolve organically and sort itself out?

ANDREW SJOQUIST: That's a good question. I heard earlier about the coordination authority that's been formed, and I too am not certain around the benefits that that's providing. I think clarity on future direction, in terms of policy, is perhaps what's missing. I think there's some concern that decision-making may be made in a shifting sands type of environment. We've compared Singapore earlier. The better comparison may be a country like Malaysia. They've had significant growth in data centres over the past two years, and the reason I use them as a comparison is that they have more sovereign-generated energy than, say, Singapore does. My understanding is that Singapore doesn't generate any or much, if any, of their own electricity. That's one of their big constraints. But if we look at Malaysia as an example, they've got lots of energy resources, as does Australia as well.

The process of converting that energy resource into data centre output at speed is really that's the thing that needs to be considered. At the same time we also need to look after the communities that we operate in. It's not just go hell for leather on making sure that we service the AI tokens or whatever it might be—there's more to life than generating AI tokens—but, I think, some certainty around the position. I think that comes from location. I think it can be done just by making particular zones, as was done with the REZs. REZs were built or architected to provide clarity around where renewable energy investment should happen. Perhaps there should be a concept of something like a DEZ, where the data centres are more welcomed and perhaps more streamlined through the approval process but in particular places and particular locations that are well geared to support them.

The Hon. MARK BUTTIGIEG: Just going back to that latency fallacy that you pointed out, is there a point in time where it does become an issue in terms of distance, because we heard on evidence at our last hearing that, in theory—leaving aside the debate on the local economic benefits and the cost-benefit analysis of building data centres—if you had enough submarine cables, you wouldn't have to build any data centres because you could just say what's the difference if it's in Singapore or wherever and we just get the data from there? At the end of the day, it's all on the cloud. We just transmit the data via submarine cables. Is there a point where, for the distances, the latency become an issue and therefore you have to build it locally?

ANDREW SJOQUIST: It depends on the application. Again, it's kind of horses for courses. In terms of AI, the latency, I believe, is less important, certainly from a training perspective. You've got two parts of AI. You've got training and inferencing. Inferencing, as well, I don't believe is that much of an issue, or it's diminishing in terms of its issue. But it's probably more of an issue than it is in training. But I think these assets that have been built around submarine cable systems and others—we should consider them an export pipe perhaps rather than an import pipe. There is a genuine opportunity here for New South Wales and Australia to use their resources in a responsible way to create a new export industry here. That export then, in turn, provides economic benefit to the State to do other social good and community-boosting projects too.

Yes, we could import all of our AI from other countries. Other countries are having the same problem in terms of building it fast enough, so this, I think, is the unique opportunity for New South Wales to participate and to be the exporter of AI technology. We're already doing it in places like Malaysia. If you look at a lot of the facilities that are being built up there, they're being built by Australian companies, and they're doing them very well. So I think there's no reason why we shouldn't, with good, solid policy and direction over the next 20 or 30 years, be able to do that back in Australia, in my view.

The CHAIR: I think that might be all we have time for. This has been an incredibly informative session. To the extent that there were questions taken on notice or there are supplementary questions coming, the Committee secretariat will be in touch. That concludes our evidence with you today. Thank you very much.

(The witness withdrew.)

(Short adjournment)

Dr ROB NICHOLLS, Senior Research Associate, Centre for AI, Trust and Governance, University of Sydney, affirmed and examined

Professor DEANNA D'ALESSANDRO, Professor and Director, Net Zero Institute, University of Sydney, sworn and examined

The CHAIR: Welcome back. We now welcome our next set of witnesses. Would you like to start by making a short opening statement?

ROB NICHOLLS: Yes, I'll start. Chair and honourable members, thank you for the opportunity to give evidence today. As I've mentioned, my name is Rob Nicholls. I'm in the Centre for AI, Trust and Governance at the University of Sydney. With me is Professor Deanna D'Alessandro, director of the Net Zero Institute at the uni. Deanna will also provide some opening remarks. The university approaches complex, contested problems like the impact of data centre governance on a multidisciplinary and interdisciplinary basis. Our appearance together reflects that Deanna is a chemist and an academic leader. She's devoted her career to tackling challenging societal problems using her STEM expertise. My first degree is in communications engineering and my doctorate is in regulation. Between us, we span material science, energy systems, law and regulatory design. The Net Zero Institute and the Centre for AI, Trust and Governance together bring that range of expertise to bear on what are genuinely wicked problems.

My focus is on term of reference (h). Honourable members will have read our joint submission. However, I'd just like to bring up some areas where the landscape has changed in important ways since we made that submission. There's more inference happening at the edge. AI compute is increasingly distributed rather than solely concentrated in very large data centre facilities. Specialised processors, particularly tensor processing units and application-specific integrated circuits, are emerging as energy-efficient alternatives to GPU clusters for inference workloads.

The move towards agentic AI means a move back to CPUs to the same extent as GPUs. Fuel cell technologies, including solid oxide systems from companies like Bloom Energy, change the step function demand on the grid. These developments don't resolve the challenges that the Committee is examining, but they do shift the problem. In my field, they change the shape of what good regulatory settings need to be. The changes create new risks but new opportunities on the path to sustainable data centre infrastructure. In my view, data centre governance cannot be addressed by any single discipline or profession. We're here to assist the Committee in whatever way is most useful.

DEANNA D'ALESSANDRO: Thank you very much for the opportunity to speak with you today. I'm Professor Deanna D'Alessandro, as Dr Nicholls mentioned, from the Net Zero Institute at the University of Sydney. As Dr Nicholls mentioned, our submission takes an integrated approach, bringing together technical policy and systems-level perspective. From our perspective, data centres are becoming defining infrastructure for both the energy transition but also long-term economic competitiveness, yet the systems that they rely on continue to be planned and governed largely in isolation. The central issue is how data centres are integrated into the system in a way that's sustainable over time. Infrastructure delivery is moving faster than system integration. While the approvals are progressing, energy systems, water systems and planning frameworks are not always keeping pace. This creates risks that are difficult to roll back later, particularly in relation to emissions, resource use and community outcomes. Without deliberate coordination, short-term decisions risk locking us into inefficient system outcomes and higher emissions trajectories.

At the same time, I think we all realise this is a huge opportunity for us all. Many of the challenges that this Committee is examining, including energy integration, cooling system efficiency and infrastructure coordination, are very active areas of research, development and innovation. Across the university, as Dr Nicholls has already noted, we're working on solutions such as system-level energy, cooling and compute optimisation, digital twins and intelligent infrastructure platforms, demand response and grid integration, low carbon and circular design integration, independent standards and monitoring, and workforce and skills development for data centre and AI infrastructure. But, of course, we realise that these solutions need integration into planning and regulatory frameworks. This means collaboration between researchers, industry and government, and real-world environments where they can be tested and scaled.

We've just had the recent strategic review of R&D, or SERD, released. This reinforces to all of us the importance of this collective effort. For this reason, we emphasise embedding research and development as part of delivery. In practice, this means precinct-scale approaches, aligning approvals with infrastructure readiness, and piloting integrated solutions in real-world settings, including through digital twins. New South Wales can and should position itself as a destination for efficient data centres by supporting local and green computing research and commercialisation. Government therefore needs to act not only as a regulator but as an orchestrator of system

integration. New South Wales should treat data centres as sovereign digital infrastructure, not simply private investment, given their role underpinning critical services such as health, education and business, amongst others. Thank you, and I look forward to the discussion.

The CHAIR: I want to talk about the issue of sovereignty and sovereign data. We hear it a lot in terms of data centre proponents saying this is one of the reasons why we need to have these data centres here. I think it's safe to say the public feels that data such as our own private information—health records, things like that—should absolutely be controlled and be the property of government. But how does that work when you've got data centres that are owned and run by overseas companies? We keep getting told that the data is fluid in lots of different places. Can you describe to us what you mean by sovereignty over our data and which data you think we should have sovereignty over?

ROB NICHOLLS: There are a couple of different aspects of what is an insightful question—that is, a difficult one. You've got some issues that we've already addressed in the pre-AI approach. That is to say, we need cloud services—like Azure from Microsoft, AWS and, to a lesser extent, Google—to be sovereign, where the data stays onshore. Each of those businesses, to a certain extent, can actually promise, "If you pay more, the data stays onshore. We lose some of that data fluidity, which is an important part of giving higher availability, so you pay a little bit more for data to stay onshore." Another example is in the telecommunications space—Commonwealth regulated, but the telecoms industry security reforms of 2018 said all of the data associated with your and my phone information has to stay onshore.

It meant that the data stayed onshore. It didn't quite mean that call centres stayed onshore, but it meant that, when call centres looked at your data, they were actually looking at a picture served to them overseas from Australia and the data didn't leave Australia. I mentioned in my opening this shift to inference at the edge. The big learning of foundation AI models is not very likely to occur in Australia. We're not looking at sovereign foundation models—or I'm not. I know others may be looking at it, but to have inference locally does two things. One, it means that data can stay onshore and stay sovereign. Two—and this is the commercial reason why businesses like OpenAI and Anthropic want this—is that the response looks quicker. It's just a matter of physics.

If I'm expecting an output from ChatGPT being served out of California, I have to send data across the Pacific; the data has to come back. No matter how fast the inferencing is, there's going to be a delay caused by the time it takes for light in a fibre optic to go two ways across the Pacific. Actually shifting to have data locally and inference locally is a logical next step in AI deployment and gives the opportunity to have a degree of sovereignty, partly because you choose to be sovereign by design and partly because there's a better consumer experience of local data storage.

The CHAIR: When you say "degree of sovereignty", what does that mean? There's the two issues there. There's the wanting to have data here, and we can argue about how much quicker it would be and whether it's worth it, but that's one of the arguments. And then the other argument is that somehow for national security reasons or other reasons we need to be able to have ownership of and control over our own data. To what extent is that actually true if you have companies from overseas being in charge and holding the keys to data centres because it's their property or having that data being hosted by Microsoft? I don't know who hosts, but basically those kinds of players who are actually not Australian companies first and foremost.

ROB NICHOLLS: I think perhaps the answer is defence and security services rely on Azure from Microsoft and AWS from Amazon. It's a trust but verify approach, which says, "Yes, we trust you, but we're also going to test. We want a right of audit. We will probably encrypt whatever we put on the local data centres, because you do still have the key to the data centre, but we need that level of security." On a more commercial basis, one of the big issues—and Australia's sort of in the middle—is what is the degree of trust that you have in AI businesses. If you don't have a high level of trust, you're not going to buy their services. That's a commercial imperative for the same businesses.

We need to build up trust. If we end up with data being used, stored locally, then shipped back over to—and it doesn't matter whether the answer is the US or China, the answer is still, "We can't trust people with our data." Ensuring that there is that level of trust says, "We'll host it locally. We'll promise to host it locally and keep it in accordance with local laws, both the Commonwealth and, depending on who the customer is, State privacy laws. We promise to do that." Because of that promise, and this is us saying, "You should trust us," the reputational harm that would come from no longer being trustworthy is very problematic.

The CHAIR: Sure. But the world we live in at the moment is rapidly changing. A lot of norms are being thrown out of the window. If we found ourselves in a war scenario, for instance, what assurances do we have that the data that we currently have stored and hosted by these global companies, which have their headquarters in other countries—what assurance do we have that we have any sovereignty over that data?

ROB NICHOLLS: It's a Commonwealth issue rather than a State issue, but the Security of Critical Infrastructure Act 2018 actually provides who are—it lists 11 forms of critical infrastructure. Data centres are one of those 11. In the event of an emergency, there is a step-in right for the Commonwealth to deal with the data held by those data centres. It's not something you want to exercise, but it is something that is actually available.

The CHAIR: But if you were a bad-faith actor in the US or wherever who decided, "We're about to have a war with Australia. Microsoft, Google and Amazon, go and make a copy of every bit of data that they have so that we've got all the information on them that we need," there is not much we can do about that, is there?

ROB NICHOLLS: No, there isn't much we can do about it.

The CHAIR: So there's a limit to our sovereignty over our own data.

ROB NICHOLLS: There's a limit to the sovereignty over our land as well. If the same actor decided that invading Tasmania was an objective, there's not much we could do there either.

The CHAIR: I'm just trying to test the limits of what the industry means when it talks about the benefits of sovereignty. Why don't we just have the Commonwealth build and own their own data centres?

ROB NICHOLLS: The major issue there is a practical one. If I wanted to build a data centre, I'm going to be wanting to do that because I'm hosting processing, whether it's graphical processing or computer processing. I am going to be a long way down the list of potential customers from NVIDIA for graphical processing unit chips, Google for tensor processing unit chips or Intel and AMD for CPUs. I'm actually not going to be the most attractive customer because I don't have the same level of demand.

The CHAIR: With respect, it's not you; it's the Commonwealth of Australia. The country of Australia surely could organise to get some chips.

ROB NICHOLLS: We could. But we're not going to be as fast as a major buyer that is already buying significant quantities and is a known and established customer. So, yes, the Commonwealth could, but if I'm NVIDIA and I've got the choice between Microsoft, AWS, Google or the Commonwealth of Australia, who is the biggest customer? Those three are bigger than the Commonwealth of Australia.

The Hon. JACQUI MUNRO: Thank you so much for coming and for your interesting and important work. I was curious about the potential for something like community benefit schemes for suburbs and communities that are living around data centres that may need additional social licence benefits to gain that social licence in those areas. Do you have any views or an understanding of any international models that we might be able to look at, or even whether that's something that we should be bothering to think about?

DEANNA D'ALESSANDRO: Perhaps I'll offer a preliminary response and then hand to Rob. I'm responding on behalf of experts at the University of Sydney who work in community benefit. Their advice here is that regulatory and planning settings must absolutely take account of differences in the size and location of data centres and the specific community needs. They point out that there is very much a place-based reality to this as well, and a contextual reality. In their advice, they suggest that infrastructure sequencing and regional flexibility is important to consider, such that planning approvals should be better aligned with realistic timelines for grid, water and workforce capacity.

They also point out that there could be land use conflicts as well and that community benefit and social licence will be critical. Communities hosting data centres should receive tangible benefits in return. They suggest here that the framework should specify what proponents are expected to contribute, including local employment and training commitments, waste re-use where feasible, community energy benefits, local infrastructure contributions and genuine community consultation. They suggest that social licence must be earned through transparency and meaningful community benefit, and not assumed. They also point out that there is an emergency and disaster resilience angle to this as well, and that planning should consider the full range of hazards relevant to the particular community area in New South Wales, including bushfires, floods, heatwaves, local grid outages, cyber incidents and physical security risks.

ROB NICHOLLS: Everything that Deanna said, but you asked for a practical example—one of the things is that, if I'm building a data centre, I need continuity of power. How am I going to do that? Even if I'm fully grid connected, I'm going to have a very big battery as part of my data centre, and I'm likely to have a diesel generator as a real backup from there. If I have a very big battery, I've got two options to deal with, but I can do it concurrently. The first one is I'm going to say my very big battery provides grid stability, and so I'm going to sell that grid stability back to the distribution network service provider. At the moment, if I built even a fairly small data centre—a two megawatt data centre—and I had a four megawatt-hour Tesla battery, I could get \$100,000 a year for that. I could pay for the battery itself. But if I wanted social licence, I could start to say, "I tell you what, we'll call this battery a 'community battery' and we'll deal with it in the same way that community batteries are

dealt with in other areas"—that is, we'll reduce the overall electricity cost to the community. If I was building a data centre and I want social licence, if I can say, "I tell you what, I'll knock 10 per cent off your energy bills for every person who's within a kilometre of my data centre," I'd be offering that fairly quickly.

The Hon. JACQUI MUNRO: I see this in a similar vein to renewable energy projects. My understanding is that there are some international examples. Is it Norway?

ROB NICHOLLS: And Scotland.

The Hon. JACQUI MUNRO: They're quite clear schemes that are essentially regulated from a national level. Is that correct?

ROB NICHOLLS: Yes, but it's also the corporate structure that you could set up in both Norway and Scotland—and it's Scots law, not English and Welsh law—that essentially can create a sort of version of a B corp, but it's community based. You say, "All right, if this part of the land that's associated with the community is provided for—and this is in renewables—wind farms, we will do that in return for the community's energy supply coming from those wind farms. Everything else, you can sell, and we're keen for you to sell it, because actually the next door village also wants a wind farm because it wants to have free electricity. That works in a renewables environment. There's no real reason to think that it wouldn't work in a data centre environment. But that form of entity is not one that we have in Australia, currently.

The Hon. JACQUI MUNRO: Regarding distributed networks and digital twins, are you familiar with whether the New South Wales Government's Spatial Digital Twin project is still in train? Does that still exist, and are we using it for planning infrastructure like this?

ROB NICHOLLS: I don't know, I'm afraid.

DEANNA D'ALESSANDRO: I'm not aware, I'm afraid.

The Hon. JACQUI MUNRO: This is a little bit left field, but I was curious about the potential for a national security risk associated with data centres—that they could be targets of nefarious actors, whether it's cyberattacks or physical infrastructure attacks. Is this something that you're considering when you're thinking about data centre policy?

ROB NICHOLLS: The answer is yes, sort of, before 28 February, and yes with a high level of worry, when some of the first targets in the UAE were AWS data centres. The rationale behind—I don't think I can get into the mind of the IRGC, but there is a strong rationale to say it's got an American sign on the door and it's vital to the interests in the area. So yes, but in the same way that large data centres that have been used up until now are also potential targets. And it is both a cyber—and electronic more generally—and whether you need to physically hammer down the door to do something or set fire to the local substation, there are risks for data centres. But they have to date been managed both through the SOCI approach and also through the understanding of the operators of these data centres that this is a risk that causes a problem when they have service level agreements with their customers that says availability is at 99.99 something of the time.

The Hon. JACQUI MUNRO: This is a data-related question. There was some exploration of the responsibility of data centre operators, whether they can have responsibility for the types of information that's being stored, basically. How does the dark web operate, for example? It exists as a pretty shady part of the internet which obviously has a lot of what most countries would consider to be illegal trade going on. How does that interact with the data centre network globally?

ROB NICHOLLS: Most data centre operators would see that if they were found to be hosting parts of the dark web, that would be reputationally problematic.

The Hon. JACQUI MUNRO: How do you find out?

ROB NICHOLLS: With difficulty. There are two parts to the dark web. There's the dark web which is simply unindexed websites that's actually more easy to find. But there's also the traditional Tor, the onion router dark web, that's hard because the whole point of the onion router is you don't know where the other end is. So even if you said New South Wales State licensing of data centres puts an obligation on the data centre operator not to host parts of the dark web, it's pretty hard to know. I'm not sure if I've given a satisfactory answer.

The Hon. JACQUI MUNRO: A data centre obviously has customers that pay to use its services, so is it then mostly a matter of vetting customers?

ROB NICHOLLS: It is, but a part of the issue is—if I was trying to do something nefarious, I'm a bit more likely to host a small server. I'd do it at home rather than bother to try and acquire space in a data centre.

The Hon. JACQUI MUNRO: I've been thinking about the idea of a data supply chain and whether it's like a modern slavery-type approach. You can't really regulate the internet, that's a part of the problem. But are there conversations to be had around the idea of a data supply chain that jurisdictions like the EU, for example, might be looking at from a GDPR perspective?

ROB NICHOLLS: I think a little bit less in terms of GDPR than the Digital Services Act and the Digital Markets Act. It does lead to odd outcomes. From today, in the US, direct messages on Instagram will no longer be end-to-end encrypted—"Encrypted messages on Instagram? Never knew about them." But there are direct messages on Facebook Messenger; they are end-to-end encrypted. In the EU they have to stay encrypted, because there is an obligation to interoperate. If you're on Messenger and I'm on Telegram, I can send you a message and the encryption won't be degraded by going through Meta's facilities.

The Hon. JACQUI MUNRO: How does that work if you're interacting transatlantically?

ROB NICHOLLS: Transatlantic? It actually lands in Europe at some point, and it still has to interoperate in Europe. But the UK says, "That's terrible, because it could be used for sending abhorrent material, so we don't want end-to-end encryption. Where there's a break, we want to be able to get in to see if things are going wrong." Even the EU and the UK have different approaches to what you are doing in a data supply chain. One says, "For competition reasons, it needs to be fully interoperable, and that means fully encrypted," and another jurisdiction says, "That's horrible. We should be able to get in and stop the abhorrent material being sent." I think the ultimate answer is lots of people are thinking about it but end up with solutions which differ, and each differing solution is unfortunate in its own way.

The Hon. JACQUI MUNRO: Have you seen different levels of investment from data centre operators depending on those rules?

ROB NICHOLLS: Meta certainly says that it has to spend more to meet its obligations under the Digital Markets Act and the Digital Services Act in the EU. That gets to be problematic, especially when the major hyperscalers are generally in US-domiciled countries, in precisely the environment that the Chair raised, where that can be problematic.

The CHAIR: Professor D'Alessandro, you mentioned carbon aware scheduling in the submission—this idea of timing computing tasks to coincide with periods of high renewable generation. Is any data centre operator in New South Wales currently doing that, and how much difference would it make at scale?

DEANNA D'ALESSANDRO: I'm afraid I can't answer the question of if somebody is doing it at the moment. Perhaps Dr Nicholls has some ideas. But it is a genuine opportunity if data centres are running off renewable energy. Given the renewable energy zones in the State, it's actually an opportunity for the State of New South Wales to genuinely move ahead its emissions reductions targets if we can get data centres to co-locate or at least co-invest with those particular renewable zones.

ROB NICHOLLS: The REZs, the renewable energy zones themselves, were not picked randomly. They have a high level of fibre connectivity, so the potential for doing that is very high.

The Hon. SCOTT FARLOW: I think we heard that from our previous witnesses as well with respect to that. You note in your submission, of course, the competition that exists between data centres and all the other users that we could have, whether it be industrial or whether it be residential. In this regional footprint potential, why is it that data centres haven't located regionally in the past? What have been the impediments, and how could it be encouraged in terms of our current framework?

ROB NICHOLLS: Why haven't they in the past is partly because you really want lots of fibre connectivity. If you look at what used to be the Switch data centre in Pyrmont, if you look at major data centres both north and south of the city, all of the big carriers have lots of fibre to them. Also, all of the processors in those plants are the processors that come from 15 years of businesses trying to reduce the energy cost per operation of the chips in there, because big data centres needed them. Graphical processing units haven't had that optimisation for energy use, which is why they need lots more energy. So, for now, why would you want to put a data centre into a regional area? Actually, there are lots of good things. Firstly, I'm likely to have space around to be able to put up both solar and/or wind, and put my battery and my diesel generator in a place that isn't going to disturb people when I test it once a month to make sure that the diesel isn't clogged within the system. That's all positive.

The location will depend on the fibre connectivity, because you actually want at least redundant paths going in and out. But building additional fibre connectivity if you know that there's a data centre going to be built is not a major impediment. If you're going to build a data centre, let's say, 20 kays north of Orange, we know there's fibre into Orange running a spur. Well, a spur's not quite good enough. We actually have to run a ring, but

we can run a ring. We can go Orange new data centre, Bathurst, and then put new fibre in between Orange and Bathurst to provide a level of redundancy. In the time it takes to build a data centre, building the telco infrastructure around it that you need is certainly feasible.

Frankly, my major issue then is, does this location 20 kilometres north of Orange have adequate water for cooling and/or, as Deanna said, some mechanism that I could resell the heat to? Actually, if that happens, maybe I'm going to look—rather than north of Orange, I might look at Blayney, because I can sell that excess heat to a pet food factory, because it actually, at the moment, uses electricity to generate heat. If I can sell high-pressure steam to a pet food factory then I'm going to actually end up—their output is cooled water, which I can then recycle back into my plants. So you start to get, as long as you think about it holistically—and we've tried to outline in our submission how to think about these things holistically. The reason there's a major impediment to building data centres in regional Australia is just we haven't done it today.

The Hon. SCOTT FARLOW: But it's been done globally.

ROB NICHOLLS: Yes, absolutely.

The Hon. SCOTT FARLOW: I think we've heard that evidence in terms of the US and Virginia, effectively the data centre capital of the US. It's not all clustered around the suburbs around DC or the like; it's northern Virginia.

ROB NICHOLLS: There's a bit too much nimbyism around there.

The Hon. SCOTT FARLOW: The Arlington types. When you have a look at that, what sort of models has the US adopted to encourage that—or has it just been organic?

ROB NICHOLLS: To the greatest extent, it's been organic. The rationale is precisely that nimbyism that you get if you try to build it just on the outskirts of Arlington. Big data centres with their own renewable backup require quite a lot of space. If you're not going to have renewable backup, you're going to get much bigger objections from locals, at least in the US. They might say, "Just a minute—my electricity bill is going to go up or my water bill is going to go up because we've got a data centre," as opposed to building in a more remote area, where there might only be 20 people who are actually providing personnel to the data centre but that's 20 more people that are going through the cafe in the local town. It's more opportunities for employment during the construction phase. It just changes the dynamics a lot.

The CHAIR: In September 2024, just as AirTrunk announced their plans for a pretty massive expansion following the acquisition by Blackstone, the Minister for Climate Change was told by the department that data centre development on its current trajectory imperils our legislated emissions reduction targets. There has been a lot of talk about the speed at which the renewables rollout is happening and whether data centres imperil us getting to our climate targets on time. Well, on time would be 20 years ago. What is your view on that and whether renewables investment can keep up with data centre rollout?

DEANNA D'ALESSANDRO: In our submission we definitely make a point of the sequencing, so to speak. As Dr Nicholls touched on, there has to be this symbiotic relationship between the infrastructure that exists and the data centre. At the minute one is outpacing the other; they're not particularly step in step. Really, the nub of the question that you're getting to is around the pacing of renewables rollout versus data centre. I think we could look at the other side of it and actually say that data centres could be a really positive way to actually enable the renewables rollout such that Australia can actually reach its decarbonisation goals. For example, if there's regulation around the amount of renewables that a data centre needs to use—that they not just build those renewables for themselves but actually they support a local community, which improves job prospects, workforce prospects and the ecosystem of the community itself—then actually it could be a real enabler if we coordinate.

This really comes to the centre of our submission, which is around whose job is it to coordinate this. On the one hand, it's the private sector. At the minute we're hearing some terminology around this being a bit of a bonanza time for people investing in data centres. It's going so fast that the rest of the infrastructure, including regulation and not just physical infrastructure but all of the governance infrastructure, just cannot keep pace with it. Partly it's a pacing issue, and this is where the question of whose responsibility is it—we feel there is a genuine responsibility of government to orchestrate this pacing, if you like.

The CHAIR: Is there a world then where we say, "Look, the speed at which data centre rollout is happening is faster than the renewables rollout and we need to put a halt on the non-critical data centre expansion until renewables has caught up?"

DEANNA D'ALESSANDRO: I think there's a genuine risk to that. We all understand, as you pointed out, Chair, that the world is a very different place. It's moving very fast, and if we do not keep pace with the rest of the world then actually it will be a genuine problem for us. I don't think we can simply turn the tap off, so to

speaking. I think it's actually on us to harness all of this expertise that we've got, which again comes to part of the recommendation here that we genuinely bring in all of this expertise, as you are, and address the question of what do we need to do. Do we need to address the broader governance issue such that we can actually figure out how to step in pace with what is happening in the private sector and data centre rollout?

ROB NICHOLLS: Just in the same line, it might also be saying, rather than halting data centre rollout, a data centre rollout which is close to a REZ—yes, you can do that instantly. But a data centre rollout that you want to do in Waterloo is going to take a lot longer to be approved. So you give signals. It's not a perfect solution, but you give signals that say, "Do your data centre rollout in a way that will actually keep us on track for our emissions targets." That may well be that you build your data centres in regional parts of New South Wales. We don't want to stop you building data centres. We just want to nudge you to the places where your rollout is going to be consistent with our—

The CHAIR: How do we weigh up the tension there? It's not a zero-sum game. We don't have abundance. Even if data centres build their own renewable energy centres, that doesn't come from nowhere. That means that we're not building those renewable energy projects for the existing demand. At what point do we weigh that up as a society and say, "No, climate change is happening and we need to get to net zero. We don't necessarily need our Netflix to be one microsecond quicker"? How do we balance that?

DEANNA D'ALESSANDRO: Perhaps I can start. I think this is really where we come to the orchestration. We need to be better connected. The interconnectedness between portfolios, departments—we can have that conversation. Everyone has a role to play here but, ultimately, somebody has to orchestrate the connectivity. That is a potential role that the State Government should play in orchestrating that and actually reconciling exactly that—figuring out what the priorities are. As Dr Nicholls points out, there is very much a place-based aspect to this, but we don't want to detract investor confidence. I think that's a common issue that we have. We can't stop this but we can reconcile how to move forward.

The Hon. EMILY SUVAAL: Similar to what my colleague was asking about net zero and the renewable energy rollout, but perhaps asking the question slightly differently, when you look at your value chain, which you very helpfully put in your submission, what role could AI and data centres play in enabling us—I think those were your words, Dr Nicholls—to reach some of these aspirations quicker? Is there a world in which that is the case?

DEANNA D'ALESSANDRO: I think that's a very genuine possibility. The reason I am optimistic in a non-pollyanna way about this—

The Hon. EMILY SUVAAL: I'm always optimistic. You've got to be.

DEANNA D'ALESSANDRO: —me too—is because we do have scenario planning that demonstrates how to do this. Coming back to the very first comment that you made, Chair, we're in a very uncertain world. Let's face it: The scenarios that now we're testing are much different to the ones that we were, maybe, 12 months ago.

The Hon. EMILY SUVAAL: So how do we do that? How do we get to that?

DEANNA D'ALESSANDRO: This is where I think the role of universities comes into this—the role of researchers. We have researchers who are developing the digital twins and the methodologies to be able to actually scenario plan under different circumstances. To me, this is where the interconnectedness between the different sectors that are part of the solution here need to come together. We have them in New South Wales. We're already assembled. I saw on the website, having read a number of the submissions to this inquiry, you had submissions and I could see solutions there from many different sectors across university, industry, government. One of the interesting benefits of working with the university sector is that we are apolitical. We come with the evidence. The data centres can actually be a very helpful proponent for reaching net zero, as much as it is yet another massive demand on our system, if we harness it and actually use it well. I'm afraid that means right here, right now, because, as you say, it is happening.

The Hon. EMILY SUVAAL: Or yesterday.

DEANNA D'ALESSANDRO: Yes, it needs to have happened yesterday. I think we are very well positioned.

ROB NICHOLLS: I'd agree. It's not just the University of Sydney; you've talked to people from UTS. Actually, what you've got within New South Wales is a range of universities that actually play nice and can bring expertise from different areas to address this problem. The apolitical nature of the work that we do is really important and can really assist here.

The Hon. EMILY SUVAAL: What would you say the suggestions are, from a regulatory perspective, Dr Nicholls, because that's one of the capacities, at least, that we have at the State level that we could possibly look at to better ensure that we're getting the community benefits that we need but also not missing this window of opportunity. What would you say, from your experience, in that regulatory framework, needs to change as a top priority in order to better or more easily enable this?

ROB NICHOLLS: One of the ways in which regulation can work is to facilitate the orchestration that Deanna has already mentioned. It's a two-way obligation, but an obligation on potential data centre constructors and the Government to be able to work in a way that provides the best outcomes. That might well be issues like focusing on where we can build our centres that is going to assist in meeting our environmental targets—proposing data centres be closer to REZs than to the city. I don't think this is rocket science.

The Hon. EMILY SUVAAL: No, it's not at all.

ROB NICHOLLS: It's merely saying regulate for the blindingly obvious and you'll get good outcomes. Having said that, of course, traditionally, that is the hardest thing to do.

The Hon. JACQUI MUNRO: Science and technology policy is not rocket science.

The Hon. EMILY SUVAAL: No, it's not at all. I live in the Hunter Valley—

ROB NICHOLLS: Yes. You're close to a REZ.

The Hon. EMILY SUVAAL: —and I see amazing opportunities, where I live, for this sort of technology. We keep talking about the future jobs that we, as a State, can have, but there is a world in which this technology can exist alongside some of the technology that's been in the Hunter for decades. We've obviously got an aluminium smelter at Tomago. We've got a number of mines. I understand some of them are looking at that sort of technology to exist in partnership, as they are equally looking at renewable energy and other things. Are you aware of any other examples, whether globally or in other States, where they have introduced, whether it's through the planning system or another way, an ideal framework by which we can enable or ensure at least that we don't see some of the issues that we've seen in some of the more historical industries, in terms of legacy impacts and otherwise, to really make sure we can make the most of the opportunities?

ROB NICHOLLS: I will give you an example that is an odd one, but hopefully it brings things together—which is Woodside. Woodside Petroleum built a digital twin and used that digital twin to start to see, "If we took this path, what would the outcome be? And if we take that path, what would the outcome be?" Using the Hunter as an example, work with creating a digital twin of existing industries and get the University of Newcastle involved.

The Hon. EMILY SUVAAL: I'm sure they'd love to.

ROB NICHOLLS: Of course, and we'll help if we can, to start to build—actually, we can do the what if, and what would normally be the regulatory failures that required an extra regulatory step, model it first. We've got the capability within this State to do that type of modelling really well. When I mentioned Woodside, I thought, what's Western Australia or the Northern Territory got to do with it? That was all done at the University of New South Wales. This is not technology or approaches where we have to go off overseas. It's all here. We can do these things.

The CHAIR: That concludes our session with you. Thank you so much for your submission and for your evidence today. It has been incredibly useful. To the extent there were questions taken on notice or supplementary questions, the Committee secretariat will be in touch. Thank you.

(The witnesses withdrew.)

Mr JEREMY GILL, Head of Policy, Committee for Sydney, affirmed and examined

The CHAIR: I welcome our next witness. Thank you for your patience. Have you got a short opening statement?

JEREMY GILL: I do, yes. Chair and Committee members, thank you for the opportunity to present at this parliamentary inquiry. My name is Jeremy Gill, head of policy at the Committee for Sydney. We are an independent urban policy think tank who advocates for the whole of Sydney for a more liveable, productive, resilient and equitable city. We welcome this inquiry. It arrives at exactly the right moment before the trajectory is locked in, and while there's still the opportunity to shape it. Sydney is the epicentre of Australia's data centre boom, and significantly global in terms of data centre investment activity. This is not a trend; it's a structural shift.

Data centres are critical economic infrastructure. They will underpin the industries that will define Australia's future and Sydney's within that—AI, advanced analytics, fintech, biomed tech, life sciences and quantum computing, among others. We also know that they're competing with essential economic land uses and industrial facilities and some innovation districts. They're reckoning with the challenges around water and energy supply, and there's an increasing community backlash to them. This paints a picture of a contemporary, systemic challenge that many cities and regions across the world are grappling with.

Our submission outlines six challenges that we believe this inquiry can contend with: that Sydney is a city and New South Wales is a State with no spatial strategy for data centre development; that the current planning frameworks they're operating within were not designed with data centres in mind; that the increasing social licence challenges facing data centre development highlight the need for them to be seen as generative, rather than extractive, in the eyes of communities; that energy and water are already binding constraints, but ones that can also drive innovative solutions; that the speed at which the sector and technology is evolving means that it's important to set performance expectations, rather than specifying technical solutions; and that the long term is planned for now, with consideration given to asset obsolescence and adaptive re-use.

Our submission is underpinned by three key messages. The first is that this is a question of how we should provide for data centres, not a question of if we should provide for data centres. The second is that we must hasten with care in the planning and the rollout of those across Sydney and New South Wales. The third is that we should think about data centres as infrastructure, not just employment land uses, and plan for them accordingly. I would be happy to answer any questions that the Committee has.

The Hon. MARK BUTTIGIEG: I'm not quite sure if this is the right witness to be asking, but I'll ask my question anyway, in the event that it's okay. It's a general question about the endgame for this industry and, as you pointed out, the timeliness of the inquiry and the burgeoning trajectory and all the rest of it. I'm curious, from an engineering point of view, do you have any insight into where this all ends? Presumably, at one end of the spectrum, there's endless appetite for exponential processing power to satisfy everyone's insatiable appetite for processing information and all the rest of it. I guess it's a function of data storage, right? Let's say we get to a point five years down the track where we've got a million data centres. Is that then enough? Is there a point at which it just plateaus and we don't need any more because we're getting rid of the old data and we've got enough capacity to keep processing what we need? Has anyone thought of what that future looks like?

JEREMY GILL: In terms of what the future looks like, I can't talk to the technical demand side in any depth. We're at the beginning of this, in relative terms, so I think that there is still globally significant unmet demand for compute power, storage power and processing power as we advance the AI-led revolution that we're seeing across the global economy. I suspect there will be a period at which there is more in the market globally and you start to see a plateauing. But I think, really, we're at the cusp of a complete transformation of how we do things with AI, so to give any certainty to how much is enough from a data processing and capacity perspective, I wouldn't like to comment with any sort of clarity because I think it's still probably too early to tell and I'm probably not the right one to get into the weeds of what that looks like over global trends.

The Hon. MARK BUTTIGIEG: No. Is there any insight as to the ratio of storage required for legacy storage as opposed to new processing power going forward? Do we have any feeling of how much capacity is used for basically storing historical processing as opposed to new processing?

JEREMY GILL: I can't answer that one, unfortunately. I don't have that level of insight into the technical side of it, unfortunately.

The Hon. MARK BUTTIGIEG: I'm sorry, I misread the schedule. It's actually a question probably better directed to the next witnesses, but I thought I'd better ask in case you had some insight. Thanks for that. I appreciate it.

The CHAIR: Your submission warns that data centres are "systematically displacing other permitted uses from industrial zones, particularly knowledge, economy and R&D tenants in innovation precincts like Macquarie Park", which you note generates "approximately \$13 billion in annual economic output". You recommend creating a new separate land use zone specifically for data centres, which was raised by the last witness as well. Has the New South Wales Government given you any indication of willingness to pursue that kind of reform?

JEREMY GILL: I haven't heard anything at this point. That doesn't mean to say that they're not exploring it. I'm not sure. The point raised there is that data centres are located in areas where we have high levels of fibre, as the previous witnesses talked about. I guess the opportunity cost, particularly in places of innovation districts, is that there is a risk that it can displace the actual act of innovation—the research institutes, the universities and businesses that are going in there—if we're not planning accordingly. Our suggestion about thinking about this as a separate land use comes partly from that.

But also, this is an asset class in an industrial zone, which is the logical place for it in the current planning controls. But, because of the price that they can pay for land, it's nothing we've seen really in industrial land use markets in the past. It's starting to out-compete other uses around freight, logistics, manufacturing and others that tend to go into industrial zones. They're also the things that we need to keep the city functioning. That's where the recommendation around thinking about it as a distinct land use zone to reflect that, actually, at the moment, they're out-competing—on a value basis and that they can pay for land—other uses we also need distributed around the city.

The CHAIR: It's also the uncertainty, isn't it? I've been out to the location where there's going to be a new data centre on Mars Road in Lane Cove, and residents looking at where that's being proposed. At the moment, it's like a small business park. People are used to having just small businesses operating out of there with houses across the road. It just seems very urban. But then they're now being faced with this prospect of that business park and those jobs being displaced and instead having what they view as quite a potentially noisy and really heavy industrial thing with backup diesel generators and all sorts of things being put there. It has come as a complete surprise to them. If we were to adopt this approach that you're talking about, then it becomes much more clear that, no, that is zoned for this particular type of use.

JEREMY GILL: There are probably two elements to that. Firstly, there's the fact that those small business parks that are distributed throughout Sydney are home to a number of the businesses that keep parts of the economy going and a whole range of other activities that don't necessarily locate in a CBD or a retail high street and need those business parks. There is a local jobs element, but there's also just a small business supply chain necessity to those precincts. They still need to be planned for and looked after. The element around having data centres in specialised zones in different locations around Sydney or around the State also is a signal to utility providers and councils and others who can then start to plan their infrastructure investment and understand the role that they might need to play in terms of servicing energy, water, utility infrastructure, rather than those industries coming into existing areas that play a whole range of different roles but also that energy, water and changing dynamic of local communities.

The reality is those industrial precincts are always going to be a place where the economy is evolving in. Like any kind of industrial or commercial enterprise area, new businesses come and go, and new land uses come and go. The value of an industrial zoning is that it permits different industries to grow and evolve by being more flexible than a commercial floor space can provide for in a business park or a CBD. You do need flexibility. You need to expect that they will evolve, but you need to also recognise that the businesses that were in them need to be supported in terms of the land values that they have.

The CHAIR: I don't think there's any substitute for going out and looking at a proposal and hearing from residents as to all of the different things that they're worried about that it's going to impact. One of the ones that I hadn't really thought much about was they were saying they have a bus route that goes up that street because there are 200 jobs or whatever currently in that business park. They're very concerned that if those jobs go from that bit, then it won't be long before bus routes are cut because there's not enough patronage. Is that something that you've looked at from a policy perspective? Is that something you've explicitly considered in terms of that flow-on effect to having these relatively jobless industrial uses in these zones?

JEREMY GILL: Yes, not to the level of the bus. That's a really interesting kind of insight into that. As industrial precincts, particularly in existing areas, do start to see more data centres, the jobs will still be there. They'll look differently; they will still be employment-generating uses, potentially not as much. They play a different economic role. That flow-on into how community services need to be addressed is an interesting one. The nature of industrial zoning is they tend to be relatively lower job density than other enterprise and commercial areas just by virtue of the space needed to do whatever the business is doing. They are always relatively lower job density.

If there is a displacement issue, this speaks to the need for a wider approach to thinking about industrial lands in the future of Sydney's economy in that if businesses are displaced, even not just by data centres but by other industries that can potentially pay more for the land in an industrial zone, you need to ensure that there are places for those businesses who are being displaced to be displaced into. That's why it's really important that we think about the industrial network in Sydney as a network, and not "Everything can go west." Certain things can go west, but we do still need the industrial zones through Alexandria, through Lane Cove, through Macquarie Park, through Cumberland to support a lot of those businesses and to enable those businesses that might be pushed out by higher value uses to still find a home because if they go too far, particularly those small businesses, often it breaks their supply chains. You start to have unintended economic impacts when those businesses aren't plugging into their business community.

The Hon. JACQUI MUNRO: We'll go back to you, but on that idea of community benefits sharing and the way that we classify land, is there not a fairly clean solution that involves a shift in land tax associated with data centres?

JEREMY GILL: Potentially. I think the challenge also just comes down to how much any business can pay for the land if they're competing on a comparable basis to other uses that can be within that permitted zone. An industrial land recalibrates the value of that land within a permitted zone, and it assumes that everyone can pay about the same for a plot of land. If one player can all of a sudden pay a lot more, it starts to price them out. I haven't done any analysis on the land tax implication of that, but you'd need to consider anything like that in terms of does it actually ensure that there's still an equal market within those zones.

The Hon. SCOTT FARLOW: If I can pick up on that, in a sense, your suggestion of a specific zoning for data centres would capture that because it wouldn't be involved in industrial commercial lands; it would be its own class, which you expect would be significantly higher than what a general commercial rating would be.

JEREMY GILL: That's right. There would then be competition on the level of who can pay the same amount for the same type of land. It could be data centres; it could be other uses that might also be able to do that. It could be things like community batteries or larger scale batteries. Co-locating those in a particular area is also valuable and they may be able to pay the same. I haven't done any analysis on that, but that's the logic behind the proposal.

The Hon. JACQUI MUNRO: So, basically, the up-front ability of a data centre operator to pay more for land needs to be addressed before you even think about the longer term redistribution through land tax question.

JEREMY GILL: Yes, I think so, just by virtue of the global capital that's sitting behind it. It's a signal of demand. This is why it is so important that we think about how we do it, not if we do it. We need to be doing it, but we also need to be making sure that it's working for other parts of the economy as well. It is going to drive AI and a whole range of other things. The place where your groceries are distributed from or the place where your car is repaired or the place that's doing some small-scale business also need to be accounted for across a city. That's why we keep coming back to we need to have a spatial strategy and a spatial way of understanding the role that these play in Sydney, but also across the State.

The Hon. SCOTT FARLOW: To pick up on that point in terms of that spatial strategy, I have got to say I am attracted to the classification in terms of either a zoning for data centres or a zoning for similar infrastructure items. We have a shortage of industrial lands, just as we have a shortage of residential land, and there's the industrial lands preservation strategy. Have you done any work in terms of identifying where these potential data centre zones or the like could be located throughout Sydney?

JEREMY GILL: Not specific analysis. The questions probably come down to what they need to be located here. We heard from the previous witnesses around fibre connectivity and the like. It's understanding what their operational and locational characteristics are, but also where does land use conflict potentially raise its head, and where can you account for land use conflict and remove it, and then also where are you getting that high-capacity energy in particular. Is it connecting straight into Transgrid or is it through the distribution network providers? It's understanding where their key assets are where that might make that easier. They're the sort of considerations we'd want to consider—so things that they might conflict with and things that you might co-exist with. They're the sort of things I'd be considering.

It's not to say that you'd have one; it could be that they're distributed in different ways across the city. Something we raised in our submission is thinking about these as infrastructure, as distinct from employment-generating uses. There is an increasing social licence question that particularly Western Sydney councils are grappling with. They've been planning for a future where they're getting hundreds of thousands of jobs moving west and redistributing that across the city. Data centres contribute to that and enable that, but there is a social licence risk that they're seeing it as data centres are coming instead of the jobs that we're promising.

That's the other overlay I'd be considering, about how are they complementing other job-creating opportunities, as distinct from just being single-use zones.

The Hon. SCOTT FARLOW: Just to pick up on your other report as well, in terms of Sydney as a renewable energy zone, I imagine, as we were discussing with our previous witness in terms of co-location with REZs, there's that potential that exists within Sydney as well when it comes to rooftop solar and particularly large-scale batteries and the like. Have you done any work in terms of how the two can match up, in a sense, and what that footprint might look like in Sydney?

JEREMY GILL: Yes. The way that we frame it in our submission is that, if done well, we could use data centres to accelerate the rollout of renewable energy infrastructure. In summary, our report that we launched last year identified that Sydney could generate up to 75 per cent of its own energy use through deployment of rooftop solar on large-scale warehouses and residential rooftops.

The Hon. JACQUI MUNRO: Hear, hear!

JEREMY GILL: The question then is how you distribute it—how you store it and how you distribute it, and at what sort of scale. If data centres can be the thing that either directly or enable through investment, development contributions or similar frameworks, the deployment of solar, not just on them but on the rooftops around on warehouses but on the rooftops of residential areas around the city, it starts to accelerate something that's already accelerated really strongly in Sydney that then talks to community batteries, large-scale batteries. The beauty of that, as well, is that the transmission infrastructure to get that generation exists. We've got capacity within those systems already, so it's not about having to bring new transmission infrastructure from the regions into the city.

Using it as a way of incentivising either through development contributions is one mechanism, or the way that power-purchase agreements might be structured in delivery of consent to incentivise renewable energy. But then I think the final part of that is how do you co-locate with batteries? How do you ensure that it's part of that broader story that we are delivering batteries at scale? What we're finding in our work with this is there's no single answer to that. What we actually need to be doing is piloting and experimenting, because there's probably going to be a whole range of ways that this can be done. How do we start to work with the next round of data centres and other major deliverers to trial how that might be done, either onsite or within the broader metropolitan system?

The Hon. SCOTT FARLOW: To that point, how do we get the community benefit of this? We've had different suggestions—cheaper energy prices, for instance, being one of them, and better internet connectivity in certain areas—but how do you necessarily package that up and actually deliver that to the communities surrounding data centres? What would you suggest as a potential radius, so to speak, in terms of where that benefit should be shared?

JEREMY GILL: The question probably depends on where they start to cluster. If you do have data centres in existing areas—Ultimo, for instance, or Macquarie Park, where you have residents and businesses already—the community contribution is different. Previous witnesses talked about residual heat being used. That could be pumped to heat the local pool. Or in a greenfield area where you might be having a new community where you're looking at recycled water and circular economy water systems. How can data centres be part of that story so that they're helping to build circular water or non-potable recycled water systems? In other areas it might be that there aren't any residents nearby. The question is how in an industrial area you might incentivise businesses to completely spec out their rooftops and enter into agreements so that excess energy that's being generated is being directed right into that data centre or a battery nearby.

I think it's probably not a one-size-fits-all—it probably depends on different parts of the city—but being clear on understanding, probably at a council level, what do the communities need and then how can data centres can deliver on that? I think that probably comes back to a broader conversation that's in the media right now about the classification of essential infrastructure. Things like swimming pools and things like community facilities are not classed as essential infrastructure. If we start to class them as that, then the development contributions for data centres, for other developments et cetera can start to be directed into that in certain ways. That will help build a broader social licence part of—particularly in greenfield communities in south-western and north-western Sydney.

The Hon. JACQUI MUNRO: I have a question about accurate demand forecasting because that was something that came up in the first public hearing we did. There was contention about how that's calculated and how it's reported as well and what government is asking for. Do you have any idea on how possible is accurate demand forecasting?

JEREMY GILL: It's probably a question best directed to people like the distribution network service providers, who are often grappling with that. I think it often is people looking at varying degrees of what they can get on a site. I probably can't comment on how accurate that is. It's probably a question for the Endeavours and

the Ausgrids of the world to understand how they're going through it, because that's something that they're grappling with.

The CHAIR: Thank you so much for your appearance today and for your excellent submission. To the extent there are questions taken on notice or supplementary questions—which I already have a bunch—the Committee secretariat will be in touch.

(The witness withdrew.)

Mr PETER DERBYSHIRE, Deputy Chief Executive Officer and Director of Policy and International Affairs, Australian Academy of Technological Sciences and Engineering, before the Committee via videoconference, affirmed and examined

Professor KEN BALDWIN, Fellow, Australian Academy of Technological Sciences and Engineering, and Energy Expert, Australian National University, before the Committee via videoconference, affirmed and examined

The CHAIR: Mr Derbyshire, you're appearing via videoconference from outside of New South Wales. Please note that for witnesses appearing from other States or Territories the normal protections under parliamentary privilege are more limited. I advise you to be careful about the comments you make during your evidence today and to avoid making adverse reflections about others. Do you have an opening statement that you'd like to give to read out to us?

PETER DERBYSHIRE: Good morning, members. We would first like to take a moment to acknowledge the traditional custodians of the land we are meeting on today, both in the room on Gadigal land and from where I am joining you today, Ngunnawal, and reflect on their long history of achievement as Australia's first scientists and technologists. We pay our respects to Elders past and present. The world we live in is increasingly driven by data. Growing adoption of artificial intelligence systems, industry 4.0 and an increasingly interconnected online environment have led to around 100 data centres in New South Wales and 270 Australia wide, the eighth most in the world. A stable political environment, excellent connectivity to south-east Asia and a strong local digital economy help make Australia a preferred destination, and New South Wales is leading the way.

Local data centres, being local users, have lower response latencies, making data access faster, and allow for our data to be stored domestically, reducing privacy and security risks to users. It is estimated that the economic value of Australian data centres could reach \$9.5 billion by 2021. But local concerns over access to utilities, noise and pollution have led to a growing international backlash. A poorly managed data centre industry could see energy bills increase, water resources being depleted and local communities disrupted. We've seen this in local communities across the United States.

An effectively coordinated industry could see local energy projects supported, businesses enhanced and strong community support. That's why ATSE's submission argued for a statewide data centre strategy supported by strong approval requirements for energy, water, noise and pollution. These will enable New South Wales to develop data centres where community impact will be minimised and where utility access will not be a burden. It will allow data centres to help fuel our energy transition to cheap, low-emissions technologies. ATSE wants to see New South Wales become a long-term data centre powerhouse. A statewide data centre strategy that maintains community support will help make that possible.

The Hon. JACQUI MUNRO: Thank you so much for your submission and for appearing today. I first want to pick up on your recommendation 1, which was about the data centre strategy, and whether you had any engagement from the New South Wales Government about the development of an apparent strategy?

PETER DERBYSHIRE: I can jump in on that one. We haven't had any direct engagement with the New South Wales Government on developing a data strategy. Our submission was pulled together based on the expertise of our fellowship and what they see as being a requirement. But we are more than happy, of course, to work with New South Wales Government on developing any such strategy.

The Hon. JACQUI MUNRO: It seems like they need some help, because they said one is coming but we are not quite sure when it is. I'm glad that you're of the same mind there. Onto more technological questions, are you familiar with how much research is occurring into improving the efficiency of data centres when it comes to water usage energy, not just in academia but also by industry? We hear a lot that research, for example, is not being commercialised very effectively in Australia, and we have really high-quality research. How do we link that research with industry? Also, how do we recognise the research that is already taking place within industrial entities?

KEN BALDWIN: I think this is a historic and ongoing issue with Australia's commercialisation of research and the need for industry to lift its game in terms of its own research capability. This is nothing new. I think that, in the case of data centres, what we find is that Australia is a bit of a taker in many respects. There are large companies from overseas that want to establish data centres in Australia. They will do most of their research overseas. They may tap into local Australian expertise but, effectively, we're really a bit of a taker in that regard. But that's not to say that we don't have things to contribute, particularly on the energy efficiency side of things. There are organisations like RACE for 2030 and others that are focused very heavily on these types of issues. They also have things to contribute as far as data centre efficiency is concerned.

But let me just make one point: Data centres will undoubtedly increase the demand for electricity over time. We see predictions by AEMO saying this will be a 7 per cent contribution to the domestic electricity load and the national electricity market by 2030, and that will rise to around 12 per cent by 2040. That is not insignificant, but let's put this into context: We will have to double, if not triple, the amount of electricity that we generate in Australia if we're going to fully decarbonise all energy use in all sectors of the economy. Increases in the order of 10 per cent or even 20 per cent will be a contributor to that, but it's not the dominant contribution. Electric vehicles will far outweigh the increased electricity demand from data centres, for example. Let's put that in context. It's not to say we shouldn't be doing something about efficiency of energy use and also efficiency of water use in data centres, but let's put that into context.

The Hon. JACQUI MUNRO: In fact, that AEMO figure was disputed by the Data Centres Australia evidence that we heard last week. Part of that is related to the information that is collected relating to the volume of proposed energy use—that it is at the maximum level rather than an expected or anticipated level. I just asked a previous question, I'm not sure if you heard it, of Mr Jeremy Gill about the availability of specificity, I guess, when it comes to—how much can data centres truly estimate how much electricity they're going to use, as opposed to a maximum?

KEN BALDWIN: Yes, and this going to be an economic and a social issue as much as it is a technological issue. What is Australia's comparative advantage in hosting data centres? Sure, we're a reliable country with a very high technological capability. We are well linked in through optical fibre cables to South East Asia and Asia more generally, so we do have comparative advantages. But as we see with the shifting geopolitical state around the world, there are many things that can come in and influence investment decisions, so geopolitics will also play a role.

Australia is a trusted trading partner with many countries. It has a stable governance system, by and large, compared to many countries, so that's an advantage when it comes to big investments such as data centres. Understanding those elements, which is part of what research does—researchers outside ATSE in other learned academies in the social and human sciences understand these types of issues. It'll be these influences that determine the figures that are projected for data centre investment going forward, as much as the technological side of things will determine that as well, I think, such as latency and data rates and the rest of it.

PETER DERBYSHIRE: To add to that, there's also the opportunity, as we invest more in data centres and develop these plans that require considerations for energy and water use and the like, that there will be an incentive for industry to start investing in areas where they can start reducing those burdens. At some point, there will be a cost benefit to them developing new ways to make their data centres more energy efficient, more water efficient, rather than having to develop their own renewable energy or water resources. I think it's a matter of making sure that we're preparing for the worst but, at the same time, incentivising ways in which we can capitalise on this opportunity without putting too much strain on the system.

The Hon. JACQUI MUNRO: Would you say then that the difference between an accurate demand forecast versus a maximum demand forecast is a bit moot?

KEN BALDWIN: I would say that they both depend on how long you think a piece of string is. One of the other factors we have to consider here is the development of the energy transition in different countries. Australia has an enormous comparative advantage in terms of the availability of renewable energy resources—solar and wind primarily—and this may not be an advantage that has yet materialised into being an attractor for data centres, but it may in the future.

For example, if you're in a country with a very high population density and you don't have the availability of land to provide additional solar and wind generation and capability to cope with the growth of data centres above and beyond that which you require for your own electrification of transport, electrification of industry et cetera, you may well wish to locate data centres in a country like Australia, which does have the land available to do these things. But that will only become apparent once the electrification of all energy use in these other countries approaches 50, 60, 70, 80, 90 per cent, whereas at the moment, that's not the case. What we might find is that in 10 years time there may be much larger demand for data centre investment in Australia simply because we're able to provide renewable energy and the land on which that is based at a much cheaper level than other countries with a high population density.

The Hon. JACQUI MUNRO: I have two more questions. They relate to nuclear-powered data centres and whether this is being usefully explored in the US. I understand Microsoft and Amazon have started engaging in this—whether it was Three Mile Island or in California—and that there is an opportunity to use nuclear power. It seems to me that small modular reactors don't really exist in a useful form at this point, but that there is investment in trying to solve that problem to be able to use SMRs for data centres. Is that something that you can speak on?

KEN BALDWIN: Certainly. The choice of electricity source for data centres will come down to mainly two factors, one being price and the other being reliability. Let's think about what we mean by reliability. Reliability obviously in the very first instance means you're going to have the guaranteed provision of electricity at whatever time of day and whatever day of the year that is required to operate the centre. But it also relates to the flexibility of the need for electricity. For example, one of the significant advantages of data centres being present in an electricity system is that they provide a significant load that is not necessarily completely constant, but it's certainly there in the middle of the day when now what we're seeing is reduced loads in the middle of the day. This is an issue for electricity networks because when solar is going gangbusters in the middle of the day, there is very little need for any other form of generation, so they tend to get curtailed. This can be coal-fired power stations, gas, wind—you name it. But the presence of very large loads in the middle of the day reduces this curtailment.

Equally, you can shift that load so that when it comes to the peak times of the day, let's say in the early evening, if you have the ability to turn down the demand for electricity in a data centre and shift it into the middle of the day, that is a significant advantage. Having flexible generation that allows that to happen is a really important factor. Now, sure, it's wonderful if you've got your own electricity source that supplies a data centre and does everything you need, but the grid is always your friend. Having a grid connection that enables input when required—and maybe it's only in cases where the off-grid source of electricity that the data centre might employ needs downtime or suddenly fails. The grid is always your friend, so having flexibility in the demand from a data centre is going to be crucial. When they make these investments in their own electricity generators, they will also be looking for flexibility because you never know when you might have to use the grid.

Are nuclear power stations flexible? Potentially. Particularly, as you mentioned, with small modular reactors, you can ramp up and down an individual small modular reactor to some degree. You could also switch one of them off in a bank of small modular reactors and dump the load somewhere else. So there is the ability to allow these generators, like small modular reactors, to follow flexibly a changing demand curve for the data centre where that might be appropriate. The same is true, though, of renewables. If you've invested in solar and wind farms to power your data centre and you have a battery system or an off-river pumped hydro system where you store the energy, that is flexible as well. Decisions will be made based on price, primarily. We all know that solar and wind with batteries is the cheapest, but it also might be based on other things like the flexibility required et cetera.

Just to make the point, there is a lot of noise around the use of small modular reactors or even full-scale nuclear power stations to supply data centres. But there's also a lot of interest and, indeed, investment by very large companies in looking at nuclear fusion, which isn't here yet. Nuclear fission is what we have with uranium-powered nuclear power stations. Nuclear fusion is the fusing together of hydrogen isotopes. That's not with us yet, but it may be in the next few decades. There are many companies investing in small-scale fusion reactors at the moment—I would suggest in the dozens around the world—and one of the big drivers is for data centres. If you had a fusion reactor that powered a data centre, it would be like having a nuclear fission reactor powering a data centre. The ability to ramp up and down might not be as good as with a battery, but it still might be possible. Let's, again, put this in context. Price will be the main determinant. Flexibility will be important. But there are many different ways of providing flexibility at a cheap price, and renewables do that for the moment.

The Hon. JACQUI MUNRO: That was very helpful.

The CHAIR: Your submission, which is excellent, argues that data centre electricity demand is near continuous and that it can help maintain grid reliability by raising minimum demand. You also suggest that AI workloads can be time-shifted to use curtailed midday solar. The Oxford Economics Report for AEMO suggests that demand flexibility in data centres is "not typically inherent" and is much more limited in co-location facilities than hyperscale ones. How confident are you that data centres will actually provide grid benefits, rather than simply adding to peak demand?

KEN BALDWIN: If you are looking to switch off a load source at peak hours in the early evening and turn it on only in the middle of the day, data centres won't have that flexibility, but they will have some flexibility. With language learning models that require learning processes to go on in the background, but that is not tied to immediate demand for that computing infrastructure, those sorts of things can be done at times when the load on the grid is not so demanding and can be shifted to the middle of the day. But it depends on what the usage of the data centre is for. If it's just simply information in, information out all the time, then the ability to make that flexible is not so great. But for AI language learning models, there is some flexibility. You can ramp things up and down to some degree, but you'll never turn it off completely at peak times.

The CHAIR: Another part of your submission looks at where the data centres should ideally be located. At one point you say that they shouldn't be located near residential areas, which instinctively makes sense because

there are other things you want to use that space for, if nothing else, even if you don't think about the water and energy drain and complications and the potential pollution and noise issues. But that's not what we've been doing, and it's certainly not what the Investment Delivery Authority is proposing to facilitate. How far away, ideally, should data centres be located, and how do we then facilitate that at a larger scale?

KEN BALDWIN: This is part of the old chestnut of not factoring in negative externalities, right—so the decreased amenity for residents, the effects on the local environment et cetera. If that were done properly, then clearly this would up the price for housing data centres near residential areas. You'd have to have amelioration of noise effects et cetera. All those things obviously should be considered and taken care of. There is a significant advantage, though, in housing data centres away from urban areas where they are powered by renewable energy. That is because you can locate them in a renewable energy zone, where the marginal loss factor of transmission is much less, rather than taking the electrons all the way to the city and using a data centre there.

You get losses along the lines and also you're not putting a burden on the distribution network. You can tap straight into the transmission network in a renewable energy zone. There are advantages of locating data centres in renewable energy zones. Apart from the fact that they're not impacting on the urban environment, there are price advantages and there are investment advantages as far as the network and the distribution systems are concerned, because they don't have to make the same amount of investment, and they don't have to pay for the losses along the transmission lines.

PETER DERBYSHIRE: Another potential option is, of course, looking at areas where fossil fuel infrastructure will eventually close down in the coming years. These are areas that have a community that is already going to be looking for jobs. They are areas connected with the power grid. They are areas of industrial development already. These are opportunities where we can place data centres that are actually going to be probably less impactful than the renewable energy infrastructure that was already there, while making use of infrastructure that they need. So there are opportunities to put these things away from residential areas while still actually helping local communities.

The CHAIR: I think the Hon. Emily Suvaal and I are on a unity ticket when it comes to the Hunter renewable energy zone being co-located with some sort of data centre zone. One of the issues with renewable energy zones is that you've got quite a large construction workforce, which leaves behind very few jobs, and there's an influx and an outflow of workers from some of these regions. If it was properly sequenced, we could move certain workers across to be also building the next data centre before they go on to build something else. Is it realistic that we could get the workforce to those renewable energy zones further out that are not of the generic construction jobs but are the more specific, as I understand it, engineers and construction personnel that need to put the chips in and things? Is that realistic for us?

KEN BALDWIN: I'm not an expert in that area, but I would guess that is possible. Let's also think about how the timescales for the energy transition look going forward. This is a process that's going to take us, let's say, up until 2040 to basically finalise the electrification of demand in all areas. This will mean continuous building of renewable energy systems, storage systems and all of the ancillary frequency control systems for the next 15 years or so. By that time there will be some of the pre-existing facilities that were built 10 years ago that will need to be refurbished and rebuilt. This is a bit like painting the Sydney Harbour Bridge. This process is not going to be a one-off that will then stop, given the time frame for the energy transition to take place not just in the electricity sector but decarbonising all the other sectors. So we won't necessarily see workforces disappear completely. They might go up and down in number in a particular renewable energy zone, but I think that, as the transition to zero carbon energy sources in other sectors of the economy goes ahead, we will have to keep building out these renewable energy zones until we decarbonise industry, transport, agriculture et cetera.

The CHAIR: True.

The Hon. JACQUI MUNRO: I had one question about education and training. Obviously it is part of your submission that we need to have a skilled workforce to support both the construction and the operation of data centres. We know we have a huge shortage of IT workers and construction workers, as it were. Are there any education and training institutions that are really doing a good job at promoting the value of IT skills? How do we get more IT workers, in particular related to your role in ATSE? I guess construction also works with engineering. How do we get more people in these industries?

PETER DERBYSHIRE: One of the key areas to get more people into these industries, as you put it, is to accept the fact that we are now working in a more global marketplace when it comes to skills. In the past, Australia has relied heavily on bringing in IT skills, especially from other countries. Our software engineering workforce—our IT workforce—has been largely brought in from overseas. The reality is that other countries where we have taken these skills from now need them themselves. It is growing globally, so we have to focus on the fact that we will have to train our own workforce.

In the IT industry especially, that means looking at areas where we are going to have opportunities to tap into skilled workers that we have largely left out in the past. I'm talking mostly about women in IT. We know that the IT industry is largely male dominated. Getting more men into IT, we're probably tapping a resource that is already overutilised. But there are going to be large numbers of women who are interested in IT who have the opportunity to learn, train and be supported in this. ATSE runs a program across the country that helps improve diversity of women in areas such as IT at universities, called Elevate. What that does is successfully support women to attain university degrees in these areas. Because it is wraparound support, not just scholarship dollars, it is support for mentoring and it is support for workplace health and safety and psychosocial welfare. We see a retainment rate of 96 per cent of women in these areas. These are opportunities where it will take time. We have to be prepared for that.

We also have to accept the fact that we're going to have to have a highly mobile workforce. This goes to the previous question as well. We're going to have to accept that people are going to have to move across States to have a mobile workforce, in order to support all of the jobs that we need to do. That comes with its own inherent challenges. There are obviously qualification challenges when you move between States; they are sometimes not always recognised. An opportunity there is also to create greater consistency between the States on qualifications within the workforce. Those are some of the solutions. Our universities and our VET sector are exceptional at training people like this. But we need to help them more in accessing those resources and those people that they may not necessarily have coming into the pipeline.

KEN BALDWIN: Can I add to that a couple of points? First of all, everyone is going through the energy transition at the same time. Indeed, Australia is leading the world in that regard. We're installing solar and wind at the fastest rate per capita of pretty much any country. We are going to have to rely on our own workforce to do that. Every other country will be doing the same, so we can't import them the way that we have in the past in other industries. The same, as Peter just mentioned, will be true in IT. But there are a couple of other things that we have to consider. One is that we will have to replace a large fraction of our export earnings that come from the export of fossil fuels with other forms of income, because that fossil fuel demand will eventually dry up. Having a new industry around IT, data centres and quantum technologies is one way that we can move ahead with that.

There is another imperative, not just because we're transforming our economy like everyone else. We've got to replace our major export earners, somehow, in the future, with something else. We really have to think not only about our domestic transition but about our export transition at the same time. The demand for skilled people is not going to be fulfilled by immigration or poaching of people from outside. The second point is that Australia's third or sometimes fourth highest foreign exchange earner is higher education. This has always been the case over the last few decades. There is another opportunity. If we are ahead of the curve in terms of data centres, IT and quantum technologies, we have the opportunity to educate other countries' people and build on that existing comparative advantage in higher education foreign exchange earnings. On those two points and focusing on our future exports, we really need to think about that deeply as well.

The Hon. JACQUI MUNRO: Love that. That was very beautifully said. Some 36 per cent of New South Wales services exports—it's the largest single component—is higher education. I think about 42 per cent of goods exports is thermal coal, mostly to Japan. Obviously, they've got a net zero target.

The CHAIR: Thank you so much for your submission and for your time today. To the extent there are questions on notice or supplementary questions—and I have some quite techy ones that I will be sending through to you, which were too techy for me to say—the Committee secretariat will be in touch.

(The witnesses withdrew.)

(Luncheon adjournment)

Ms ROCHELLE FLOOD, Deputy Mayor, Lane Cove Council, affirmed and examined

Mr MARK BRISBY, Director, Planning and Sustainability, Lane Cove Council, sworn and examined

The CHAIR: I welcome our next set of witnesses. Would you like to commence by making a short opening statement?

ROCHELLE FLOOD: Yes.

The CHAIR: Please go ahead.

ROCHELLE FLOOD: Thank you for the opportunity to appear today on behalf of Lane Cove Council. Council recognises the need for digital infrastructure like data centres as part of our modern economy, but we do hold significant concerns about the planning process surrounding their rapid expansion and the considerable impact they are having on our communities. Data centres aren't just another factory or standard industrial building. At the scale now being proposed, they're essentially an extractive industry. They require enormous amounts of electricity, water and land to be viable. Local communities are left to bear the costs while receiving little to no public benefit.

Lane Cove's experience shows why the current planning system is not equipped for the rapid expansion of hyperscale data centres, particularly when those data centres begin to cluster in small areas. In Lane Cove West, we are seeing a cluster of data centres emerging in a small business park that's very close to homes, the local school, childcare centres, the community nursery, significant bushland and our parks. The problem is that these proposals are still being assessed one by one, rather than looking at the cumulative impacts. We need a strategic planning framework that properly assesses clusters, precinct-wide impacts and the suitability of locations before we get to the stage of a detailed DA.

The continued use of the SSD pathway is also causing problems. There are minimal requirements for meaningful engagement with council and communities on behalf of the proponents, and this often leaves our community in the dark. We need a proper framework that gives communities a voice and councils a voice in the planning process because it's the communities who are the ones that experience the immediate impacts. In Lane Cove, we've already seen construction-related disruption from data centre infrastructure, including cabling works that caused significant damage to both public and private property. I've also been in contact with residents who are increasingly concerned about noise impacts and the potential cumulative impacts on air quality from the testing and regular use of backup diesel generators. The business community is also deeply concerned about the loss of employment land. Right now, we're looking at losing 50 per cent of our business park to data centres. Data centres create short-term construction jobs, but their ongoing local employment opportunities are minimal.

There are also broader concerns about resource use. The demand for electricity has implications for the grid, emissions reduction targets, as well as who's paying for the infrastructure. Water use is also a major concern. We know a lot of these data centres are using our drinking water for cooling, and while we may not be in a drought right now, Sydney is an incredibly drought-prone area, and this could be a concern for the future. There needs to be requirements for renewable energy and recycled water, as well as a genuine user-pays model where proponents fund the essential infrastructure they need, rather than shifting those costs onto the public. On that note, Lane Cove also supports community benefit arrangements for the local community so that there is a genuine public benefit. Finally, we would like to see a moratorium on new data centre approvals until such time as these core issues are addressed and we have a more sustainable industry that does give local communities a voice in the planning process.

The CHAIR: Can I just start by asking a few sort of basic questions. How many data centres have you got already in your local government area?

ROCHELLE FLOOD: We currently have two, and we have more in the pipeline. We're looking at up to four in the planning process at the moment.

The CHAIR: So you could have up to six. Are they all in the same sort of general part of the LGA?

ROCHELLE FLOOD: Yes. The majority are in the Lane Cove West Business Park, and then there's also one just slightly over towards Mowbray Road, which is a little bit further away but still nearby.

The CHAIR: How do people find out that these things are happening, and how do people then feel about how they can contribute to asking questions or even objecting to these developments?

ROCHELLE FLOOD: That has been a real problem. With the latest one we've had, which is the Project Mars proposal, I've heard directly from residents who live in properties that essentially adjoin the business park right where the data centre is going to go. Some of them didn't receive an initial letter of notification. They

weren't aware of the online Zoom meetings that the proponent held with the community because some of them are in their eighties and are not particularly digitally savvy, so they weren't across those things happening. They actually didn't find out about it until a neighbour doorknocked them and told them that it was up on exhibition, and this was only two weeks before submissions closed on the formal DA.

The CHAIR: This idea of social licence is kind of thrown around, but what is the impact of that lack of—or at least that feeling of lack of—consultation on the willingness of people to have these things in their neighbourhoods?

ROCHELLE FLOOD: I think it has made people very unwilling to have them, and it has made them quite concerned about the process because they really do feel like they don't have the time to get their heads around what's proposed. They sort of find out two weeks before the opportunity for them to have their say closes. There are hundreds upon hundreds of pages of documentation for them to look through and get their heads around. In the case of the most recent one, council did ask the proponent to hold a public meeting with the community much earlier on in the process, and they declined that request.

The CHAIR: Which proponent was that?

ROCHELLE FLOOD: That was Goodman.

The CHAIR: You make the very clear and unequivocal recommendation in your submission that the State Government remove data centres from the State significant development pathway entirely and return assessment to local councils, which would be a significant structural change. The Government also now has this Investment Delivery Authority. We're still a little unclear as to what it's doing, but it seems to be trying to pave the way and help, almost like a concierge for these proponents. Can you tell us why you believe that's necessary and what your experience has been of trying to influence data centre proposals once they are in that SSD pathway?

ROCHELLE FLOOD: The experience of attempting to influence them in the SSD pathway has been very difficult, from speaking with both community and staff. My understanding is that the engagement with the proponents has very much been a tick-box exercise once it's in the SSD pathway. They don't require council's formal approval, so they can essentially come and say, "Here's what we're proposing." That may not align with our LEP or any of our DCP requirements. Once they've had that engagement with council, that's a box they can tick and then off they go. That's sort of the end of it with some of them. That's causing quite a bit of frustration, and I think the community is seeing that as well.

The CHAIR: Your council resolution of 19 March this year notes concerns about "brownouts and blackouts increasing in Lane Cove West since the expansion of data centres". Can you tell us a bit more about that? That seems to be quite a real impact already of these data centres in your area.

ROCHELLE FLOOD: At this stage, anecdotally and from looking at the outages through Ausgrid, we do see more blackouts and brownouts in the Lane Cove West area. That has been the case since we've had more of the data centre developments. That's something that has been reported to us. People are getting quite frustrated by it. We've seen that correlation. We're not saying it's necessarily proven conclusively, but we know that data centres, when they switch to their backup generators, can destabilise the grid and can result in dropouts for other users when there's that sudden switch. People have been reporting more blackouts and brownouts since we've had them.

The CHAIR: The Investment Delivery Authority has endorsed another data centre project—the Lane Cove Data Centre Development Project, which I understand is at 16-20 Mars Road in Lane Cove West—to be delivered by something called the Lane Cove DC Alliance. That project seemed to have come out of nowhere in that it didn't seem to have any planning documents lodged at the time, but did when it was announced. What does council know about that one? What has your involvement been in that proposal?

ROCHELLE FLOOD: Could I defer to Mr Brisby?

MARK BRISBY: As the Chair mentioned, there wasn't a great awareness of this new authority. Council first became aware of this quite large proposed data centre in Lane Cove West through the media in the first instance. We then had to research and follow up with the State Government and the department of planning to find out a bit more. You are correct that there are no plans or details. We've had no contact from any applicant.

The CHAIR: So we have this massive new data centre that is being endorsed by the IDA without having spoken to council at all about anything?

MARK BRISBY: That's correct, yes.

The CHAIR: When I look at the history in the area of these data centres, there seems to be this rolling development of new and expanded ones, one after the other. What is the impact of roads being ripped up

repeatedly, new water connections being installed and there being something new all the time? Is that something that residents are talking to you about?

ROCHELLE FLOOD: It is, yes. We've already seen from one of the current ones, which is an AirTrunk facility, that when they were doing additional cabling works, there were significant disruptions. That caused damage to footpaths, street trees and public infrastructure. Some people's private driveways were even damaged in that process as well. It took several months and quite a bit of public pressure, and the hype of local media putting pressure on them and investigating who was behind it, before we saw any action to rectify.

The CHAIR: Was there then any sort of compensation back to council for any of that?

ROCHELLE FLOOD: Yes, that has now been fixed, but there were months—over a year, really—of disruption for the local community.

The Hon. SCOTT FARLOW: In terms of the timeline with the data centres in Lane Cove and the Lane Cove West industrial park, you've got the AirTrunk data centre and the Goodman project. When did they both first start operating?

MARK BRISBY: The original AirTrunk development commenced operation in late 2020. The Goodman one is yet to commence construction.

The Hon. SCOTT FARLOW: So it's only one that's commenced at the moment, is that right?

MARK BRISBY: That's correct.

The Hon. SCOTT FARLOW: So there's one that's operational within Lane Cove and then there's four other proposals, is that correct?

MARK BRISBY: That's correct.

The Hon. SCOTT FARLOW: And they're all clustered around Lane Cove West and, I think you said, Mowbray Road? I'm familiar with the industrial park there. I'd say it's fairly unique for an industrial park, because you've got the residential areas that aren't connected by road, or effectively have the walkway that go through them, and then the access that comes off Epping Road into the area. It's unlike other industrial complexes where maybe residents might see that there's some benefit because there aren't as many car or truck movements that go through the area. This is an area that is completely quarantined, so to speak, from the traffic impact that occurs within that commercial precinct. Is that fair to say?

MARK BRISBY: That's correct.

The Hon. SCOTT FARLOW: Except for parking, I suspect; they might have concerns about workers parking on the streets.

MARK BRISBY: It's also embedded in pristine bushland. Major waters that lead out of the harbour—Lane Cove River and Parramatta River—and adjoins low density residential, along with a local public school.

The Hon. SCOTT FARLOW: In terms of that precinct, what's been the feedback as to why that's been attractive to data centre operators?

MARK BRISBY: The anecdotal evidence we can establish is just through meetings with applicants. They tend to say to council that Lane Cove has very good central access to water and power.

The Hon. SCOTT FARLOW: You note in terms of the 15 and the City of Ryde and in Willoughby as well, you've had the ones at Artarmon and St Leonards that have been there for quite some time on the Pacific Highway; of course, we've seen in Ryde and around Macquarie Park that there's been the proliferation of data centres in that area as well. This seems to be a relatively new movement. Are they getting pushed out of some of those other areas they've been traditionally operating in, or are they finding it more difficult to acquire property in those areas and this is the new spot that has been identified?

MARK BRISBY: Again, it's anecdotal. It appears the properties are available. Obviously, in the Lane Cove area the property values are a little bit higher, but they still can facilitate that. But it seems to be the attraction of our access to water and power.

The Hon. SCOTT FARLOW: In terms of council's ambitions for that industrial area, prior to the introduction of data centres, what was council's master plan or strategy for that area, and what sort of industries would you be typically thinking would occupy those spaces that data centres have now moved into?

MARK BRISBY: The Lane Cove West industrial park, which is endorsed in the Government's current draft Sydney Plan, was a high-tech industrial park that had technology manufacturing and distribution centres,

and generally fairly high employment. The lack of employment with data centres against the potential employment by other uses is a significant loss to the community.

ROCHELLE FLOOD: If I could add to that as well, I have spoken directly with some of the local business owners in that business park. They expressed concerns about the fact that they are being pushed out under the Goodman proposal, which is the Project Mars development. In that one the proponent represented that the current occupants were not interested in staying onsite. I've been told by the business operators that that was untrue. They had recently invested, in some cases, upwards of \$20,000 in renovations to their premises because they were planning to stay there for a much longer period of time. They feel they've got a great local business community there. They're quite upset about the prospect of being pushed out. They've said that they don't think they'll be able to find something else nearby, so those are local jobs that will be lost.

The Hon. SCOTT FARLOW: The evidence we've heard as well is that when it comes to the ability to pay—as you said, Lane Cove is a higher cost area than others. But when it comes to data centres, there is a higher ability to pay. That's largely driven by the global demand when it comes to data but also perhaps by an ongoing lower employment footprint as well, so the ongoing operational costs in that sense might be less. There is that challenge for existing businesses who might be competing with data centres in such a zoned area.

The CHAIR: Do you know approximately how many permanent operational jobs the data centres in your LGA provide compared to the employment uses that they replaced? Have you got any modelling or information on that?

MARK BRISBY: Not exact modelling, because it's hard to predict future uses. The original AirTrunk that's been operating since 2020 employs half a dozen people at any one time over a 24-hour period; that's against a very large site that would have employed anywhere from upwards of 50 to 200 people.

The CHAIR: So that one is 12 Mars Road, isn't it?

MARK BRISBY: The Goodman, yes.

The CHAIR: So that business—but the one that you say there's already existing business owners there, how many jobs are at that site?

ROCHELLE FLOOD: Sorry, that's what I was trying to confirm. Is that a question I could take on notice and come back to you, please?

The CHAIR: Yes, absolutely, and how many the proponent is saying will be left afterwards in the data centre, as well, if it's built,

ROCHELLE FLOOD: I don't know if they've given us the number that they're expecting afterwards, but I can definitely come back and confirm the number that are there currently.

The Hon. EMILY SUVAAL: You've got an existing data centre within the LGA. I'm interested in what the Committee can perhaps learn from that process, albeit that was some time ago. When that data centre went through all the relevant permissions and those sorts of things, and just noting some of the concerns in your submission, particularly around power use and water use, were there requirements put in that the operator would have to offset its energy use by providing a certain amount of either solar battery or otherwise renewable energy firming power, and in terms of the water use? We've heard from data operators, at least through submissions, around their capacity to use other types of water that's not necessarily potable water. In fact, in many cases, they don't need potable water. They can use other things. I'm interested if that was put into the consent. If not, why not? How can we ensure those sorts of things are put into any future consent for data centre operators?

ROCHELLE FLOOD: My understanding is no, there was no requirement around recycled water use or offsetting around energy or anything like that. Again, in terms of the broader question you asked about learning opportunities, I think with one of the data centres that we do have—AirTrunk, for example, was a bit more willing to engage with council and the community, which was one benefit at the time. However, what we've experienced with Goodman has been completely different. The current system doesn't really require them to meaningfully engage, so it's on a case-by-case basis in terms of whether they're a good corporate citizen and if they're going to try and meaningfully engage or not. I think it needs to be tightened up in that respect to require early engagement with councils—having it sent to the designer, having an initial conversation with council, having it sent to our design review panel and then having the final design based around that. Having more genuine community input earlier on in the planning as well, through public meetings, would be ideal.

The Hon. EMILY SUVAAL: Obviously, we've got the State significant development pathway, which I understand you've got some concerns around, but I also understand that that community consultation is built into

that pathway. Are there tweaks we can make to that or are there issues with the way that currently works that would mean that councils would be more included or involved?

ROCHELLE FLOOD: I think there could definitely be tweaks made to that process that would enhance it. What I mentioned earlier was that, with the current proposal at 12 Mars Road, there were some residents who immediately adjoin the site where it's going to be developed who didn't receive notification letters and weren't at the meetings and things like that. Again, it really does need to be tightened up in that respect. If there was more opportunity for councils to be engaged, we could then also create DCPs that relate directly to data centres. Part of the issue is that they look at our DCP and say, "We've got a 10-metre setback; that's appropriate," but that's for a residential building. It's quite a different story when you're talking about a data centre that adjoins significant bushland. You need a bigger setback. But because we're not involved in the planning process, we don't have those frameworks to guide them. I think that's something that could be better embedded in the process.

The Hon. EMILY SUVAAL: Would a pathway specifically for data centres—given there's obviously a need and a demand for them—assist, do you think?

ROCHELLE FLOOD: I think that could help, yes. If there was a clear requirement for engagement with councils early in the planning stage through that pathway as well, that would be very helpful.

The CHAIR: You say in your submission that you wrote a letter to the relevant Ministers seeking certain outcomes. When was that letter sent and who did you send it to?

ROCHELLE FLOOD: That was from an April 2025 council resolution, so the letters would have been sent approximately 12 months ago, I would say.

The CHAIR: Which Ministers, and did anyone respond?

ROCHELLE FLOOD: Could we take that one on notice, please? We will come back with a response.

The CHAIR: Please do. In terms of the contribution that you would be expecting to make if there was a proper consultation process, there is this tension where people talk about, "Consultation means delay and obstruction and the chance to object." But in your experience of negotiating and being consulted on these sorts of projects, what sensible input would you be expecting to have that could lead to better outcomes? Are they things like, "Actually, when we look at the cumulative impact, we've got this water problem," or "This street is also being dug up for some other reason"? Is it simple sharing of information so that everyone is on a better footing before the project begins?

ROCHELLE FLOOD: Yes, I think that would be an important part of it, just sharing that information and having a process that does allow for the assessment of cumulative impacts in a local area, as well as what I mentioned earlier about council having a design review panel. If you are building close to sensitive bushland, if you've got light spill or concerns like that, the panel would be there to look at those things. Again, if we were working within a particular pathway and we had a specific DCP for data centres, I think that would give us a framework as well to make sure that we were looking at it in a sensible way. I think the other key thing is just the social licence aspect, because if we are involved as a council then we are able to go out to the community to talk to them earlier on. They don't feel like they are being blindsided and then they are just left worrying about what's happening and feeling like they are in a rush to get their submissions together. The only option at that late stage is to oppose it because they have had no opportunity to provide meaningful input before it goes to that stage.

The CHAIR: I guess it's also that stress of something happening to you, as opposed to having any control over what happens in your neighbourhood. Can I ask the number of backup diesel generators for that project at 12 Mars Road? I think it was 49 of them. Do you know? One of the things we were talking about earlier was this lack of cumulative impact when they're thinking about those sorts of approvals. If all of the diesel backup generators for all of the data centres in your area were to suddenly be switched on, do you know which of them have generators, how many it would suddenly be and what the impact on air quality would be? Is that something council has oversight of?

ROCHELLE FLOOD: No, it's not. We don't have any—

MARK BRISBY: No, we don't. Every data centre has diesel fuel support so they never go down in a blackout. There is a cloud of secrecy around the hazard licensing of those diesels, and that puts a lot of the community at unease at that. The secrecy around it is probably what concerns the community.

The CHAIR: Which means that you can't manage it. For example, that site—again, because I'm familiar with it because I've seen it—is diagonally opposite is a school. If you had a blackout in the middle of the day—the kids are playing in the playground—and if you knew the extent to which that air quality was going to be impacted, you could maybe put a plan in place to stop children breathing in that air. There's secrecy and then

there's causing a health hazard. Is that the sort of thing that, if you had better oversight of on the cumulative impact, you could ask for those sorts of emergency plans and things to be put in place?

ROCHELLE FLOOD: Yes, and air quality monitoring and things like that for when it's in use.

The CHAIR: Air quality monitoring is another important one.

The Hon. EMILY SUVAAL: I was going to ask a question similar. The current operation that you have, the current data centre—is that information that they share with you the use of their backup diesel generator?

ROCHELLE FLOOD: No.

The Hon. EMILY SUVAAL: There's no requirement for them to do so?

ROCHELLE FLOOD: They don't share it with the council, no.

The Hon. EMILY SUVAAL: So if they were to use it, you wouldn't know. Does anyone know how often they're being used? Because it seems to me like an important piece of the puzzle that the local area would need to know at least.

ROCHELLE FLOOD: My understanding is there is a testing regimen for the data—

The Hon. EMILY SUVAAL: Monthly.

ROCHELLE FLOOD: There's the requirement to do that, so that's a known use, I guess. But in terms of when it's used in emergency circumstances, as a council we're not informed.

MARK BRISBY: No.

The Hon. EMILY SUVAAL: It would be interesting if there were alternate—

The CHAIR: Could that be in brownouts—as well as blackouts—that they might use those?

ROCHELLE FLOOD: Yes. My understanding is that sometimes they are. If there's a high-demand period, then they can be directed to switch before there's a blackout. They can be directed to switch to using their backup source.

The CHAIR: Would council know? Would the schools know?

ROCHELLE FLOOD: No.

The CHAIR: We have run out of time. That was incredibly useful. Thank you very much. To the extent there were questions taken on notice or there are supplementary questions, which I think there might be a few, the Committee secretariat will be in touch. That concludes our time with you.

(The witnesses withdrew.)

Councillor JESS MILLER, Deputy Lord Mayor, City of Sydney Council, affirmed and examined

Ms MONICA BARONE, Chief Executive Officer, City of Sydney Council, affirmed and examined

The CHAIR: We now welcome our next witnesses. Thank you so much for appearing. Would you like to commence with a short opening statement?

JESS MILLER: Sure, thank you. I'm here today representing Lord Mayor Clover Moore, who's very sorry that she couldn't be here and sends her apologies. Thank you, firstly, for the opportunity to speak in support of the city's submission. The city's submission is based on evidence and research. It recommends that data centres be approved within a framework of strong regulation to ensure that they are operated sustainably and transparently. Australia is on track to become the second largest host of data centres globally by 2030. We acknowledge that data centres are crucial digital infrastructure and their being located in Australia, in New South Wales and in the city does offer a range of economic benefits. In New South Wales there are 90 data centres currently operating. In fact, in New South Wales, investments in data centres have soared by 65 per cent a year on average for the past three years, reaching \$2.6 billion in the year 2024-25.

In the City of Sydney there are currently 16 data centres operating. There are 15 in the State significant development pipeline, valued at close to \$30 million. The Investment Delivery Authority is assessing 26 submissions worth \$100 billion in the first round of its expression of interest process. However, we cannot continue to assess data centres based solely within an investment delivery framework. Doing so trades off local amenity, character, jobs and secure access to electricity and water for a big, impressive number that, for many everyday people, remains quite abstract. As a local elected representative, I, the lord mayor, councillors and city staff are hearing a lot of concern from the community about the impact that these quite lifeless, block-long, anonymous, cavernous buildings with no street activation are already having on communities. Under the current framework and without any regulatory guidance, communities rightfully do not see the benefit that this infrastructure delivers to them. There is currently almost zero social licence to operate within the City of Sydney.

Data centres are a very big business with very big impacts. In the next four years in New South Wales alone, they're projected to consume 11 per cent of our grid electricity and 20 per cent of our drinking water. Rapid, uncoordinated growth without proper planning will strain the electricity grid, use huge amounts of drinking water, undermine our climate goals, take up scarce land needed for housing and jobs, cause local environmental impacts and provide limited local economic benefits. Allowing data centres in areas zoned for light industry means we are forced to permit the use of scarce inner city land for data centres, which risks displacing productive, job-rich businesses, whose workers also contribute to our area economically and socially. With proper planning, consideration and thoughtful design, data centres might offer the potential to catalyse investment into much-needed renewable energy and water infrastructure for the communities where they are built.

The City of Sydney has been at the forefront of advocating for proper planning for data centres so they do not become an unacceptable drain on community resources. In December council unanimously supported a motion asking the CEO to explore ways to ensure better community and environmental outcomes from data centres, such as identifying best practice design and engineering models for data centres that ensure water and energy efficiency to minimise the use of drinking water, harness renewable energy, and enhance visual impact and amenity; and investigating the potential for developer contributions to fund water and energy sustainability programs for the wider City of Sydney area, particularly for the sites of data centres to recycle water and install solar power onsite. Recently, the lord mayor joined other capital city lord mayors in Canberra advocating for coordinated national planning and enforceable sustainability requirements for data centres. Just last month she spoke to other global city leaders at the C40 online meeting, where the City of Sydney committed to joining the C40 global sustainable urban data centres pact.

An example of why planning pathways matter is exemplified by the State-approved 90-megawatt centre on Harris St, Ultimo. That one draws down directly on baseload grid power and drinking water within our LGA. That is the equivalent to powering 145,000 homes and the water supply of 5,000 local households. This was approved prior to the introduction of a 2024 amendment to the State environmental planning policy, which excludes land within the City of Sydney LGA from these SSD triggers. This means that currently the City of Sydney is in a unique position, in that we are the only jurisdiction in New South Wales where proposed data centre developments are assessed by the city and not considered State significant and assessed through the SSD pathway.

The exemption to the SEPP recognises our capacity to assess major development. The example of a State-approved 90-megawatt data centre on Harris Street, now operating in the centre of our city, is proof for exactly why council is the appropriate assessor for what our local infrastructure can support. In the absence of any other regulatory or planning lever, there should be no change to the city's exclusion from SSD. Instead, we need

higher performance standards, incentives to innovate and design well and more transparency so that data centres contribute to local communities by way of renewable energy supply as opposed to just take from that.

The city's submission to this inquiry includes 15 recommendations, including statewide rules for assessing cumulative impacts on the energy network and water supply; requiring large centres to be grid positive—for data centres that use over 10 megawatts of energy, the city wants them to invest directly in new renewable energy; updating planning policy to require high-efficiency energy standards, such as NABERS Energy, and allowing them to invest in local renewable energy and storage; banning the use of drinking water in water-stressed areas; and establishing a register of data centres. These and other recommendations are reasonable and straightforward. We urge you to consider them and look forward to your report.

Currently, the community is rightly suspicious of data centres. The impression they have, based on what they see, is that they are ugly, intrusive and parasitic. They take more than they give. However, we also recognise that they are an inevitability. If designed well, utilising innovation, design and engineering, they may well offer an opportunity to contribute meaningfully to our communities becoming more energy and water resilient. In doing so, they can and should earn the social licence from our communities to operate. This does not necessarily mean via offsets and power-purchase agreements in other States; it means really catalysing that investment locally and investing in renewable energy infrastructure in situ, whereby the exact drawdown of a data centre on the grid where it is located is directly offset by new renewable power that the local community can immediately benefit from.

The CHAIR: The idea of a 90-megawatt data centre smack bang in the middle of our city that takes the same amount of energy as 145,000 homes is pretty shocking. Is that what you said?

JESS MILLER: Yes.

The CHAIR: What does that data centre do? Do you know?

JESS MILLER: We don't know.

MONICA BARONE: It collects data. It stores data.

JESS MILLER: It collects and stores data.

The CHAIR: Do you know why it had to be in the middle of the city? Has it ever been explained to you why that couldn't be somewhere else, rather than on prime land?

JESS MILLER: The assumption is because it is located near a transfer station. By virtue of it already being there and having access to that transfer station, it's effectively shored up. But we weren't part of that process, so we don't know.

MONICA BARONE: I think that's really such an important question. I can only guess for exactly that—because it could be guaranteed a supply of power and water, and there was nothing restricting it using that power and water. Of course, this whole conversation is about how do you determine that. I can only guess that, but I think one of the things that we're concerned about is that that is not good land use in the inner city. There are plenty of places—

The CHAIR: It doesn't pass the sniff test, does it?

MONICA BARONE: —but the power and water supply would have to then be provided in other places. Of course, that's what the deputy lord mayor is talking about: They ought to have an obligation to provide the power and water.

The CHAIR: If you're a data centre and you've got a lot of money or you've got a set amount of money, you could either buy cheaper land somewhere else and have it connected to energy, water and everything else, or you could buy more expensive land in the heart of the city that's already really well connected. So they just could afford it—is that the summation? Without the restrictions, it just made good commercial sense.

JESS MILLER: We would assume so, yes. I guess the other consideration too—and what causes us an intense amount of concern as well—is that we've got very significant housing targets for the Pymont-Ultimo strategy, where we have to deliver thousands more homes. Assumedly, those thousands more homes are also going to require energy, so we've created, inadvertently, this situation whereby the data centre's claim to that power—by virtue of being there first—is greater than that of future residents. The other question that we have is, within the context of the Bays precinct and the planned uplift over there, what cost is that additional investment into energy infrastructure going to mean for the cost of developing that land into future housing?

The CHAIR: In the submission, it talks about how you've got a net zero target by 2035 for the LGA—for the city. Do you know how that project impacts on that ability to reach net zero? Is it factored in then to your—

JESS MILLER: No.

The CHAIR: No.

MONICA BARONE: Net zero was about not using fossil fuels, so it's not about how much energy is used; it's the type of energy. If the data centre is using renewable energy, that doesn't affect the net zero target. The issue is that, I think, when you come back to this one that we were talking about, it's because there are going to be so many that you have to have this inquiry. If there was only one now and then, then maybe the normal grid and water supply would have coped, even though we don't think it's great land use in the inner city. But, because of the growth, that's why you're doing what you're doing. How are we going to manage that growth?

The CHAIR: So it doesn't have backup generators.

MONICA BARONE: Everything has backup generators. The city has backup generators.

The CHAIR: But the 16 data centres you have, do you have that same issue where they've all got these diesel backup plants?

MONICA BARONE: Assumedly. I think everyone has backup generators.

The Hon. EMILY SUVAAL: There's a second loop that goes around. The city, essentially, has a second backup loop. There's a city ring. It's an Ausgrid question.

The CHAIR: I'll come back to more targeted questions then. As you say, your council has this unique carve-out from the State significant development pathway. What has been the difference then? When you get a new data centre proposal, what have you been able to achieve? Because it has not gone through SSD but because you are able to actually consider it yourselves, what have been the kinds of conditions or things that you've been able to put onto those data centre developments?

JESS MILLER: When it comes through our planning pathway, ultimately we get to determine where the most appropriate location for that data centre would be. It also ensures that the planners within the city are part of the conversation. Where you've got an SSD pathway, it's incredibly opaque. When you've got a city pathway, we undertake the same consultative processes that we do for other types of development. In practice, it means that our preference is that the data centres are located in the southern employment lands. That is away from residential use.

The comparison between the Harris Street location and, for example, the one located on Gardeners Road in the southern employment lands is that the one in the southern employment lands isn't detracting from the character or use of Ultimo. In the southern employment lands, it's effectively an industrial use and highly appropriate, whereas this monolith on Harris Street means that there's no retail shop frontage and there's no engagement with the public. It's just this monstrous thing that nobody really can see what it is and it's not a contributory factor in any way, shape or form to the city.

MONICA BARONE: Can I just add to that? Local government can only condition things that are consistent with State plans. One of our recommendations states:

Introduce statewide amendments that mandate measurable potable-water limits, recycled-water cooling, and firmed-renewable procurement ...

We can't condition that until the State—because we can never put an environmental condition that is higher than a State policy. We can ask, but we can't make you. This is why this inquiry is so important. We're hoping you will set the new environmental—"environmental" in a broad sense—conditions that enable us then to say, "If you come, you have to do this and you have to do that." There's the water, there's the energy and there's the amenity—all those three things. We haven't got a framework to deal with that in our planning policy from the State because it's new.

The CHAIR: Lane Cove Council was just here telling us that they don't have this carve-out that you have. I'm trying to look at this recommendation as to whether councils should—it shouldn't be part of the SSD and we should at least allow councils to have a say. I'm trying to work out what the tangible differences have been that you've been able to achieve with your data centre proposals as a result. For example, do they tell you what the data centre is for or is there any more transparency when it's done through council?

JESS MILLER: I think it's less about differentiating between a tangible proposal rather than having something that enables us to set a standard. There is nothing—no lever, no guide, no framework—for us to reach for that compels a proponent to do better. Having talked to the kinds of engineering companies that do this work, it's great work for them because it's very simple. It's input in, input out. There's nothing that we can use currently, because there is no planning instrument at the State level, to compel people to do better. This is potentially a really positive thing that could be introduced because the benefit of data centres is they attract so much investment.

There is a lot of capital coming in for the design of these things. But there's nothing that we can use to compel this capital or this design to be harnessed and used to benefit the community.

Whatever the thing is that you arrive at, whether it's a framework or whether it's a planning framework—ideally, it would be a statutory instrument because it's real and not just a "nice to have". What we want to be able to do—and compel proponents to do—is be innovative and be creative. There's a real opportunity to get a triple-point score. If you want to build a data centre, invest in local renewable energy infrastructure. If you want to do a data centre, provide your own water. That could mean that you're improving the water quality of a nearby catchment, for example. If you want to improve land that has deteriorated, it's about giving us the opportunity to identify where that benefit could come from. Currently, in the absence of any of these standards, we get a very low standard; whereas, if we had a lever to pull, we could potentially incentivise these proponents to do much better. In doing so, they earn the social licence of the community. All anyone has to look at at the moment are these hideously ugly, huge boxes that have this really creepy hum.

The CHAIR: It is a creepy hum.

JESS MILLER: If we can move beyond that—and people are frightened. The most engagement on social media that I've had in the whole time that I've been in council was about letting people know how much water and energy data centres use. It went gangbusters because people were really shocked, really curious and really concerned. As much as we can do to exemplify what's good, the better our collective ability to earn social licence can be, and I genuinely think that they could be used to deliver a whole range of community benefits.

The Hon. EMILY SUVAAL: I'm interested in exploring the approval pathway a bit more. Have you approved a data centre under your own planning mechanisms? If so, what were the benefits of that, if any?

JESS MILLER: I don't think we have, but I'll take it on notice and double check with our planning team.

The Hon. EMILY SUVAAL: That has probably eliminated a lot of my future questions because I was going to try to go down that pathway if you had. Safe to say, the existing data centres—or data centre. Is it just one that you've got in the City of Sydney?

JESS MILLER: No, we've got—

MONICA BARONE: Sixteen.

The Hon. EMILY SUVAAL: So the existing data centres, as they were or are approved—as far as you know—don't have requirements around recycled water, energy offsets or anything like that.

JESS MILLER: No. I think there's a risk, too, in simply offsetting the problem, because that doesn't address the hyper-localised issue of energy requirements.

The Hon. EMILY SUVAAL: We've also heard some evidence that data centres can be useful in stabilising the grid in terms of the peak periods, trough periods and other things, but there does need to be the co-location of things like batteries, solar and all those sorts of things—albeit it may not be the case that that can be achieved within the constraints of the City of Sydney in particular because you're such a unique local government area. I'm keen to explore a bit more the concept that data centres aren't contributing to the area, and understand why you think that is the case. What does the City of Sydney do with its data, for example? Where does the council store its data?

MONICA BARONE: We store a lot on our own site and then we have data warehousing, I think, out in Western Sydney. We have a number of warehouses for data security reasons. We have a number of sites so that, should there be either a situation where we can't get into our facilities or we lose something, we've got a backup.

The Hon. EMILY SUVAAL: What are you doing to ensure that those sites are using recycled water, renewable energy—all that sort of thing?

MONICA BARONE: That's a really good question, and I'm going to—

The Hon. EMILY SUVAAL: And are they?

MONICA BARONE: I think, because the city has its own targets around those things—I would absolutely have to take that on notice to be sure, but everything that we do extends out to the contracts we have and things like that.

The Hon. EMILY SUVAAL: It would be interesting to see. I personally find it hard to accept that data centres aren't contributing in a meaningful way, because they are: We all rely on them, we all need them and we all use them. Whether we like it or not, they are a requirement and a necessity. What we're trying to get at is where

do we put them, what is the regulatory path like, and what are the constraints we need to put around them. That community benefit piece that I think you were talking about, Councillor Miller, is key.

MONICA BARONE: If I could talk about the contribution piece, if I may. Firstly, we accept that we're all using technology and, therefore, we all have data and therefore it's all going to be stored. What we've got is a situation, though, where there's a huge acceleration.

The Hon. EMILY SUVAAL: Correct.

MONICA BARONE: So we've got a huge acceleration, and our planning system has not kept pace with what's coming. That normal data storage that all of us have—you have here at Parliament, every organisation has—has just been part of the normal system, but we've got an acceleration of this need of the data centre. There are different aspects about contributing. As Councillor Miller, the Deputy Lord Mayor, said, the City of Sydney is a very dense area and it has housing targets and job targets. From a planning point of view, the way we achieve a sustainable city and better quality of life and productivity is to ensure that as many people as possible can live as close as possible to where they work. We need lots of homes, and I think about 75 per cent of our homes now are multi-unit dwellings—flats—and we need lots of floor space for people to work—lots of people. Those big buildings have 5,000 to 6,000 people in them. We need to protect that nexus between where you live and where you work so that we can not have people travelling for hours and hours to get to work. That's bad planning.

If you then use a lot of land for something that has no people in it, or one or two—I don't know how many people, but you know what I mean; it's not lots of people—that's a bad use of that land in this area. It's not saying they shouldn't exist, but you need to put them in the right place. The other thing is there is nothing worse—we all work so hard to get late-night activation and all the rest of it. The last thing you need is a whole lot of flat surfaces with nobody going in and out. It's dangerous. It's ugly. It doesn't work for a city. It should be in areas that are meant for that, and that's fine. That's one of the contributors. It doesn't contribute to urban amenity. It doesn't contribute to the nature of what a good, dense and sustainable city is. The other points we're talking about is its contribution to water and energy, so whether it contributes or at least pays its own way. If it can do more, that's great.

The Hon. EMILY SUVAAL: There was certainly evidence this morning that said that.

MONICA BARONE: But as long as it pays its own way. Generally, development has to be able to ensure that it has adequate water and energy and not just take from there. We just want to be sure that as you plan for this going forward that the mechanisms for it to pay its own way are there. You're right; some of it could happen on site. There are plenty of purified water plants that are small enough and could be on site, or they just become part of and pay for Sydney Water's purified water program. The sewer is just there. That's where the purified water comes from. They could put a certain amount of renewable energy creation on their own facility, or they might have to invest in something else off site. I think what the Deputy Lord Mayor is saying is you might get a bit more social licence if you actually went one better and not only got what you needed but did a bit more for someone else.

The Hon. EMILY SUVAAL: Community batteries are another potential.

MONICA BARONE: Exactly. Those programs around the idea of utilising passive roofing space and others to generate power to put into areas that don't have the roof space—these are all good ideas. But I think generally it's not a great inner-city land use for other planning reasons, and the acceleration is going faster than the ability to provide the water and energy that it needs. It needs to be regulated. That's all we're saying. The regulation has not kept pace with what's coming, which is why you've called for this inquiry. You know that.

The Hon. SCOTT FARLOW: With the City of Sydney, you've had data centres for quite some time. We've sort of heard in a sense that this could be like—I think the expression that was used earlier today was it's like painting the Harbour Bridge, effectively. We're going to get to this position where data centres will then be re-used and updated. You've had some experience of that. When was the first data centre in the City of Sydney?

MONICA BARONE: I'll have to take it on notice.

The Hon. SCOTT FARLOW: I'm also just interested as to you looking forward in terms of any future data centre approvals. Is there any ability for sort of mixed use when it comes to data centres? I'm thinking of that Harris Street one, for instance. You have the street front, which is just completely barren and open. I think there might be an awning over it or that. Is there any possibility for activation on the ground floor? I take your points in terms of what is best land use. But considering some of the sites where you've already got them in place, is there any possibility for coexistence, so to speak?

JESS MILLER: Yes. There are a range of precedents in other cities where you've got really interesting mixed use. These things generate a lot of heat. In some circumstances, that heat can be captured to do other things.

There are hydroponic-aquaponic setups on top of data centres. There are really interesting ways that we could make this work. To Monica's point, there is no shortage of land that is in pretty poor condition. When I consider where we've got some of the most polluted and worst water quality in Sydney, you can't really go beyond the Cooks River. Throughout that catchment, there are vast tracts of land on either side that are highly industrialised.

Were there some kind of regulatory framework to compel proponents to think outside the box, we might get some of these more interesting proposals. I think that's a good thing. But in the absence of any regulation, motivation or reason for people to think differently about it, we're not seeing any of that innovation, because it's cheaper and quicker to just do the standard black box version. It's an excellent question. There's lots of room for innovation. That would be fantastic, whether that's thinking about design excellence. That's a way that the city has been able to get really interesting outcomes for the public realm. That's a great idea. In the absence of any of that, we're just getting what we get, and everyone's getting upset.

The CHAIR: Should that regulation, then, be retrospective? For instance, to Mr Farlow's question, should we be trying to say, "Actually, you do now have this obligation to have some sort of amenities if requested"? Or even to your recommendation 6 on energy usage, should we be going and saying, "Actually, there's now a need to retrofit and you must have minimum six-star energy"? Is that something that we should be considering as well?

JESS MILLER: I think it depends on the lifecycle of the asset. To the best of my knowledge, the useful expected life is about 25 years max. I'm willing to bet that will contract as the technology improves. They might end up taking up less space. There's going to be a range of iterations. One way that we could compel that kind of improvement is through a plan of management. We can't, I don't think, necessarily change the rules once it has been approved, unless there is some kind of regulatory catalyst for doing so, which you might provide. That'd be interesting.

Also within this conversation, it's worthwhile looking into how the assets are managed, because there's an opportunity at the application point of view to compel changes in the way that it's built and the way that it operates. To the best of my knowledge, the proponent is not usually the person operating the different cells. They effectively create a warehouse, under which certain cells or aspects of the warehouse are then let out. It's also really interesting to consider supply chain and governance of these businesses to really understand where the best intervention can be made. If we're a local government and we're a customer, we can use procurement to potentially compel a better outcome. It's understanding what that process looks like and where are the various touchpoints to get an outcome.

The CHAIR: Thank you so much for giving us the benefit of your experience and for your submission. It is incredibly useful. To the extent that there are questions taken on notice or supplementary questions, which I again flag that I've already got, the Committee secretariat will be in touch. That concludes our time with you today and concludes our public hearing.

(The witnesses withdrew.)

The Committee adjourned at 15:40.